



ONLYIAS
Nothing Else

IDMP TEST SERIES - 2022

High Yielding PRELIMS TEST SERIES

Test – 24

Science & Technology/Environment revision

Topics covered:

- Environment Revision
- Science & Technology Revision
- Current Affairs August & September 2021



Scan the QR code for Schedule

DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

TEST BOOKLET SERIES

Test Code:

DPP

**TEST BOOKLET
GENERAL STUDIES
TEST - 24**

E 24

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO

TOPICS COVERD: Science and technology revision, Environment Revision, Current Affairs August to September 2021



Q.1) Recently, Indore-based Enking International in India became the world's first company that operates in the carbon markets space, consider the following Statements with reference to 'CARBON TRADING':

1. These are market-tradable instruments that are issued to entities whose projects reduce carbon dioxide emissions.
2. This carbon trading began under the Paris Agreement.
3. It allows nations which are unable to meet their reduction targets to purchase carbon credits.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.2) With reference to the 'Forum for Decarbonizing Transport', consider the following Statements:

1. NITI Aayog and World Resources Institute jointly launched the 'Forum for Decarbonizing Transport' in India.
2. The project aims at bringing down the peak level of GHG emissions in Asia.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.3) With reference to 'Hydrogen Fuel cell', consider the following Statements:

1. A fuel cell combines hydrogen and oxygen to generate an electric Current by an electrochemical process.
2. Water and heat are the only by-product in an electrochemical process.
3. It can be a cheap source of electric current.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 and 2 only

- d) All the above

Q.4) The term "Surya Putra" was seen often in news in the context of -

- a) It's an ISRO mission to explore the solar system.
- b) It's the name of an award to develop solar energy at a reasonable cost.
- c) It refers to all the nations which fall between Tropic of Cancer and Tropic of Capricorn.
- d) It is UNEP's campaign incentives to increase production solar energy.

Q.5) With reference to the 'Green – Ag: Transforming Indian Agriculture', consider the following Statements :

1. The project supports harmonization between India's agricultural and environmental sector priorities and investments.
2. It will help India achieve national and global environmental benefits and conservation of critical biodiversity and forest landscapes.
3. This project Implemented by the Ministry of Environment and Forests and Niti Ayog at national level.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.6) With reference to the 'National Mission for Sustainable Agriculture', consider the following objectives of the mission:

1. It helps to promote location specific integrated and composite farming systems.
2. To conserve natural resources through appropriate soil and moisture conservation measures
3. To adopt comprehensive soil health management practices based on soil fertility maps.
4. Promote effective management of available water resources and enhance water use efficiency.

Which of the Statements given above are correct?



- a) 1, 2 and 3 only
- b) 2, 3 and 4 only
- c) 1, 3 and 4 only
- d) All the above

Q.7) With reference to 'Green Charcoal ', consider the following Statements:

1. Green Charcoal Hackathon launched by the Ministry of Environment and Forests.
2. The main objective is to clean the air by eliminating farm fire, producing renewable energy out of the agro residue.
3. It helps promote local entrepreneurship, and to increase the income of the farmers.

Which of the Statements given above is/ are correct ?

- a) 1, 2 and 3 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.8) Which of the following is /are benefits of "Torrefied Biomass" over untreated biomass?

1. Higher calorific value
2. More homogeneous product
3. Higher bulk density
4. Excellent grindability
5. Durability
6. Hydrophobic nature

Select the correct answer using the code given below:

- a) 1,2,5 only
- b) 1, 3, 6 only
- c) 1, 2, 3 and 6 only
- d) All the above

Q.9) Recently, SWAS, STAR, SAFAL Sometimes seen in news is related to -

- a) Name of planets recently discovered
- b) New technologies to convert waste into wealth
- c) It is eco-friendly green crackers
- d) It is medical treatment to cancer

Q.10) With reference to 'Integrated Farming System', consider the following Statements:

1. Employment opportunity.
2. Higher productivity.
3. Augmented returns and recycling of organics.
4. Control the built-up of insects-pest, diseases.
5. Better utilisation of land and labour.

Which of the above is/are can benefit/ benefits the Integrated Farming System?

- a) 1, 3 and 5 only
- b) 2, 3, 4 and 5 only
- c) 1, 2, 4 and 5 only
- d) All the above

Q.11) With reference to ' The United Nations Convention to Combat Desertification ', consider the following Statements:

1. It is the legally binding international agreement on sustainable land management.
2. It has a bottom-up approach and encourages the participation of local people in combating desertification.
3. The UNCCD is one of the agreements among the three among the three Rio Conventions.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.12) Consider the following Statements, with reference to ' TRAFFIC: THE WILDLIFE TRADE MONITORING NETWORK':

1. It was launched by the ministry of environment and forests.
2. It aims to ensure that trade in wild plants and animals is not a threat to the conservation of nature.
3. It supports local and national economies and helps to motivate commitments to the conservation of wild species and their habitats.

Which of the statements given above is/ are correct?

- (a) 1 and 2 only



- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.13) With reference to the ' CONVENTION ON THE CONSERVATION OF MIGRATORY SPECIES ', consider the following Statements:

1. It is an intergovernmental treaty under the United Nations Environment Programme.
2. India has not signed this convention.
3. The only global convention specializing in the conservation of migratory species, their habitats and migration routes.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.14) With reference to 'NATIONAL BAMBOO MISSION', consider the following Statements :

1. The National Bamboo Mission is a Centrally Sponsored Scheme with 100% contribution from the Central Government.
2. It is implemented by the ministry of Environment.
3. It aims to promote the growth of the bamboo sector through an area-based regionally differentiated approach.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.15) Consider the following pairs with reference to conventions and their purposes :

Conventions	Purposes
1. Stockholm Convention	— Persistent Organic Pollutants
2. Basel Convention	— Protection of the Ozone Layer

3. Minamata Convention

— Protect human health and the environment from anthropogenic emissions of Mercury.

Which of the above pairs is/are correctly matched?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1 and 3 only

Q.16) In which one of the following States is the ' Pobitora Wildlife Sanctuary 'located ?

- a) West Bengal
- b) Arunachal Pradesh
- c) Assam
- d) Mizoram

Q.17) Recently, Kasarkod and Padubidri beaches in Karnataka were accorded 'Blue Flag' tag , consider the following Statements with reference to ' Blue Flag ' :

1. This certification is awarded annually by the ministry of environment and forests.
2. A 'Blue Flag' beach is an eco-tourism model to provide tourists with a clean and hygienic safe and healthy environment.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.18) Recently, ' PARIVESH ' term was seen in news related to

- a) Afforestation strategy with public participation.
- b) Local rain harvesting system in Rajasthan.
- c) A Single-Window Integrated Environmental Management System.
- d) Primary education system for tribal people.



Q.19) Consider the following Statements with reference to , ' India Water Resources Information System':

- 1.This is a web portal containing information related to Water Resources.
2. It was launched by the ministry of environment and forests .
3. The National Water Informatics Centre is responsible for maintaining and updating the India Water Resources Information System.

Which of the Statements given above is / are incorrect ?

- a) 1 and 2only
- b) 3 only
- c) 2 only
- d) 1 and 3only

Q.20) With reference to "Paramagnetism" , consider the following Statements :

1. It is a property of few materials which are weakly attracted to an applied magnetic field.
2. It is the result of an unpaired electron having a magnetic dipole moment.
3. Titanium and aluminum are metallic elements that are paramagnetic.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3 only

Q.21) With reference to the 'Zero liquid discharge (ZLD)', consider the following Statements :

1. It is an engineering approach to water treatment where contaminants in water are reduced to solid waste.
2. ZLD technology includes pre-treatment and evaporation of the industrial effluent.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2

- d) Neither 1 nor 2

Q.22) Recently ' KAZI 106F 'was often seen in news related to

- a) It is a helicopter in the Indian Air Force.
- b) It's a genetically modified rice variety.
- c) It is **India's only Golden Tiger** found in **Kaziranga National Park**.
- d) It's the name of a planet discovered by ISRO.

Q.23) Which of the following States where India's first snow leopard conservation centre will be established ?

- a) Sikkim
- b) Himachal Pradesh
- c) Uttarakhand
- d) Jammu and Kashmir

Q.24) 'Joint Forest Management (JFM)' is a landmark initiative in the history of Indian forestry. However, its impact remains limited due to which of the following reasons ?

1. Bureaucratic Centralization.
2. Inequitable benefit sharing.
3. Participation of local communities.
4. Conflict with Panchayats.

Select the correct answer using the code given below :

- a) 1, 3, and 4 only
- b) 2, 3 and 4 only
- c) 1, 2 and 3 only
- d) 1, 2 and 4 only

Q.25) Consider the following statements about state of environment report, 2021:

1. It is an annual publication by the UNEP.
2. Tarapur in Maharashtra emerged as the most polluted cluster.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.26) With reference to koala, sometime in the news, consider the following statements:

1. It is an African animal, often called the koala bear.
2. Koalas rely on the eucalyptus tree for both habitat and food.
3. They are listed as vulnerable by the IUCN Red list.

Which of the statements given above is) are correct?

- a) 1 and 2 only
- b) 2 Only
- c) 2 and 3 only
- d) 3 only

Q.27) Consider the following statements about sea snot:

1. It is formed when algae are overloaded with nutrients.
2. Recently, for the first time in Turkey sea snot has been recorded.
3. It has caused mass deaths among the fish population, and also killed other aquatic organisms.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.28) With reference to ambergris, consider the following statements:

1. It is a solid waxy substance that floats around the surface of the water body and at times settles on the coast.
2. Ambergris is so valuable it is referred to as floating gold.
3. It is used in the perfume market, especially to create fragrances like musk.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.29) Consider the following statements about positron excess phenomenon:

1. The positron access phenomenon is the behaviour of a positron below 10 GeV.
2. It is the behaviour of a positron above 100 GeV.
3. It is the behaviour of positrons between 10 to 300 GeV.

Which of the statements given above is/ are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 3 only

Q.30) Which of the following are base quantities as per the SI (international systems of units)?

1. Mass
2. Charge
3. Length
4. Ampere

Select the correct answer using codes given below:

- a) 1,2 and 3 only
- b) 1 and 3 only
- c) 2 and 4 only
- d) 1,3 and 4 only

Q.31) Consider the following statements about Ebola virus disease:

1. Ebola virus disease was first discovered in 1976 near the Ebola River in what is now the Democratic Republic of Congo.
2. It is a zoonotic disease and spreads in the human population through human-to-human transmission.
3. There is no vaccine to prevent Ebola virus disease.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 1 only
- d) 3 only

Q.32) With reference to the 'PUSA decomposer' recently seen in the news, which of the following is correct?



- a) It is a fertilizer to increase the productivity of rice.
- b) It is a fungi- based liquid solution for stubble burning
- c) It is a weed killer.
- d) None of the above

Q.33) Consider the following statements about neutrinos:

- 1. The neutrino is a tiny elementary particle, but it is not part of the atom.
- 2. They cannot be produced in the lab.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.34) Which of the following is/are the objective of the 'Hope' mission launched by the UAE ?

- 1. To understand the climate dynamics and global weather map of Mars by studying the lower atmosphere of Mars.
- 2. To estimate the water vapour content in the atmosphere of planet Mars.
- 3. To explain how the weather of Mars affects the escape of hydrogen and oxygen.

Select the correct answer using the codes given below:

- a) 1 and 3 only
- b) 2 and 3 only
- c) 2 only
- d) 3 only

Q.35) Arktika- M satellite was launched, recently in news by-

- a) IN-SPACE
- b) NASA
- c) ISRO
- d) ROSCOMOS

Q.36) Consider the following statements about Space hurricanes:

- 1. It is a huge, funnel-like, spiral geomagnetic storm that occurs above the polar Ionosphere of Earth, during extremely quiet conditions.
- 2. They are made up of plasmas, consisting of extremely hot ionized gases that rotate at extremely high speeds.
- 3. They are related to the aurora australis phenomenon.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3

Q.37) Consider the following statement:

- 1. Coronaviruses are a group of related RNA viruses that cause diseases in only mammals.
- 2. COVID-19 transmits when people breathe in air contaminated by droplets and small airborne particles containing the virus.
- 3. The human immunodeficiency virus (HIV) is a retrovirus.
- 4. Rabies is a bacterial disease that causes inflammation of the brain in humans and other mammals.

Which of the statements given above is/are incorrect?

- a) 2 and 4 only
- b) 1 and 4 only
- c) 1, 2 and 3 only
- d) 1, 2, 3 and 4

Q.38) With the reference to cruise missiles.

- 1. A cruise missile is a guided missile used against terrestrial targets.
- 2. Cruise missiles are designed to deliver a small warhead over small distances with low precision.

Which of the statements given above is/ are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.39) Baikal-GVD, recently seen in news, It is -



- a) Giant underwater space telescope
- b) Underwater tunnel in Lake Baikal.
- c) Research Observatory in Lake Baikal.
- d) None of the above

Q.40) Consider the following statements:

1. Human in space policy for India, 2021 guidelines are applicable to those activities which shall be defined and carried out in future in conjunction with IHSP.
2. It lays the foundation for a sustained exploration in the long run that will enable presence in LEO and beyond.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.41) With reference to Nobel prize in physics, 2021 Consider the following statements:

1. Scientists have been awarded for their discoveries concerning black holes.
2. This is the first time climate scientists have been awarded the Physics Nobel.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.42) Recently 'SC120-LOX' was in the news, it is -

- a) hypersonic spacecraft
- b) Ballistic missile
- c) Liquid oxygen Tank
- d) Stun gun

Q.43) 'Epilepsy', sometime in the news, is -

- a) New research in the field of biotechnology.
- b) Recently found plant species.
- c) Neurological disorder.

- d) NASA's mission to Mars.

Q.44) In the context of covid-19 vaccination the term 'Hybrid Immunity' was seen in the news, consider the following statements regarding this :

1. It refers to the immunity which is generated by taking two different types of vaccines.
2. The immunological advantage from hybrid Immunity arises mostly from memory T cells which evolve in the lymph nodes.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.45) With reference to the Zeolites, which is the important part of the oxygen concentrator, Consider the following statements:

1. Zeolites are crystalline solid structures made of silicon, aluminum and oxygen.
2. Zeolites occur naturally but are also produced synthetically on a large scale.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.46) Recently 'nano urea(liquid)' was in news, Consider the following statements regarding this:

1. It contains nano nitrogen particles of size range (20-50 nm) dispersed in water.
2. These nano fertilizers have been developed indigenously.
3. It is slightly costlier than the conventional urea.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 and 3 Only
- c) 1 and 2 Only
- d) 1,2 and 3



Q.47) Consider following statements regarding 'Main Bhi Digital 3.0' campaign:

1. It is a special Campaign for Digital Onboarding and Training for street vendors.
2. This campaign has been launched by the Ministry of Housing and Urban Affairs(MoHUA) in collaboration with the Ministry of Electronics & Information Technology (MeitY).
3. The campaign has been launched on pilot basis in 115 aspirational districts selected by NITI aayog.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 and 3 Only
- c) 1 and 2 Only
- d) 1,2 and 3

Q.48) Consider the following statements regarding 'One Sun, One World, One Grid' (OSOWOG) Project:

- 1 The idea of OSOWOG was floated by India in the UN Climate Change Conference in Paris in 2015.
- 2 It aims for a trans-national electricity grid supplying solar power across the globe.

Which of the statements given above is/are Incorrect?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.49) Recently 'SC120-LOX' was seen in the news, it is related to:

- a) Technical platform for Supreme court to track the case records,
- b) Heaviest semi-cryogenic propellant tank made by HAL(Hindustan Aeronautics Limited),
- c) Recently developed Malaria Vaccine.
- d) NASA's satellite program to the sun.

Q.50) Consider the following statements regarding 'Lucy Mission':

- 1 This is the mission of Russia's space agency 'Roscosmos'.
- 2 Aim of this mission is to search for life on the mars

planet.

- 3 The estimated Life of this mission is 12 years.

Which of the statements given above is/are Incorrect?

- a) Only 1 and 2
- b) Only 2 and 3
- c) Only 1 and 3
- d) 1, 2, 3

Q.51) In the context of Recent testing of 'Agni-5 missile', consider the following statements:

- 1 Agni-5 capable of striking targets at ranges up to 7,500 kilometres with a very high degree of accuracy.
- 2 It uses a three-stage solid fuelled engine.
- 3 It is a Surface-to-Surface Ballistic Missile.

Which of the statements given above is/are correct?

- a) 1 only
- b) 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.52) To counter the climate change there is a shift towards the more clean energy sources and one option is hydrogen, in context of this consider the following statements:

1. Blue hydrogen is produced mainly from natural gas, using a process called steam reforming.
2. Pink hydrogen is generated through electrolysis powered by nuclear energy.
3. Yellow hydrogen is generated through electrolysis using solar power.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 and 3 Only
- c) 1 and 2 Only
- d) 1, 2 and 3

Q.53) 'Digi-Saksham' has been launched as a joint initiative of the Ministry of Labour with which MNC?

- a) Google
- b) Microsoft
- c) Amazon



d) IBM

d) None of the above statements is incorrect.

Q.54) Consider the following statements:

1. Global wind report 2021 Released by The Global Wind Energy Council (GWEC).
2. Nationally Determined Contributions (NDC) Synthesis Report released by The UN Framework Convention on Climate Change (UNFCCC).

Which of the statements given above is/are correct?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

Q.55) With reference to India's defence related developments, 'Swordfish' is related to

- a) Radar Systems
- b) Ballistic Missiles
- c) Unmanned Aerial Vehicles
- d) Nuclear powered attack submarines.

Q.56) With reference to India's Science and Technology sector the terms, Apsara, Cirus, Zerkina, Kamini, Dhruva are associated with

- a) The Earth Observation Satellites launched by the Indian Space Research Organization.
- b) Nuclear Research Reactor in India.
- c) Hybrid varieties of food crops developed by Indian Agricultural Research Institute in collaboration with the Department of Biotechnology.
- d) None of the above

Q.57) Which of the following statements is incorrect with respect to the mechanism of nuclear power generation ?

- a) Uranium undergoes nuclear fission to generate heat to generate energy.
- b) Moderator of the reactor works to sustain the fission reaction.
- c) The control rods of the reactor prevents the excessive fission reaction.

Q.58) Consider the following statements with respect to Hypergolic propellants.

1. They require no ignition source for functioning.
2. The Hypergolic propellants are not suitable for spacecraft maneuvering and related purposes.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.59) Consider the following statements with reference to Halogens:

1. Halogens can exist in free state and are strong reducing agents.
2. Fluorine, Chlorine, Bromine and Iodine are collectively known as Halogens.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.60) Which of the following statements is/are incorrect with respect to Ozone ?

- a) Ozone is formed on the upper layer of the atmosphere by the action of UV rays from the sun on oxygen.
- b) Ozone can be used as a germicide and disinfectant.
- c) The concentration of ozone formation peaks during the early hours of the morning.
- d) Both b and c are incorrect .

Q.61) Consider the following statements with respect to Metalloids:

1. Metalloids have a metallic appearance and are brittle in nature.
2. The metalloids are used in flame retardants,



pyrotechnics and semiconductors.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.62) Consider the following statements with respect to Biogas:

- 1. It is a mixture of gases produced during decay of biomass in the absence of oxygen.
- 2. Biomass is an excellent fuel and burns without producing smoke.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.63) Which of the following statements is incorrect with respect to the Atlantic Meridional Overturning Circulation ?

- a) It is a large system of ocean currents that carry warm water from the tropics northwards into the South Atlantic.
- b) It is driven by differences in temperature and salt content.
- c) As warm water flows northwards it cools and some evaporation occurs, which increases the amount of salt.
- d) If it collapsed, it would increase cooling of the Northern Hemisphere, sea level rise in the Atlantic

Q.64) Consider the following statements with reference to the Tungabhadra River:

- 1. It is a sacred river in southern India that flows through the state of Karnataka to Tamil Nadu.
- 2. It's Major tributaries are the Bhadra, the Haridra, the Vedavati, the Tunga, the Varda and the Kumdavathi.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only

- c) Both 1 and 2
- d) Neither 1 nor 2

Q.65) Consider the following statements with reference to the The National Legal Services Authority:

- 1. It has been constituted under the Legal Services Authorities Act, 1987 to provide free legal services to weaker sections of society.
- 2. It is a two tier structure at central and state level.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.66) Recently, North Eastern Region District SDG Index 2021 is released by -

- a) Ministry of Rural Development
- b) NITI Aayog
- c) Ministry of Statistics and Programme Implementation
- d) None of the above

Q.67) Consider the following statements with reference to the United Nations Convention on the Law of the Sea :

- 1. It was adopted and signed in 1982 and became effective in the year 1994 to act as a legal framework for marine and maritime activities
- 2. It is the only international convention which stipulates a framework for state jurisdiction in maritime spaces

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.68) Which of the following statements is incorrect with respect to the TAPAS (Training for Augmenting Productivity and Services ?



- a) It is a standard MOOC (Massive Open Online Course) platform with course material such as filmed lectures and e- study material
- b) It is Developed by National Institute of Social Defence (NISD) under Ministry of Skill and development.
- c) The main objective of introducing courses on social defence is to impart training for the capacity building of the participants
- d) None of the above.

Q.69) Who among the following wrote the books 'BudoAngla', 'KhirerPutul' and 'Rajkahini'?

- a) Abanindranath Tagore
- b) V. O. Chidambaram Pillai
- c) Mahatma Gandhi
- d) Dheeran Chinnamalai

Q.70) Which of the following statements is incorrect with respect to the 'Hunar Haat' ?

- a) It is an exhibition of handicrafts and traditional products made by artisans from the minority communities.
- b) The exhibition is organised by the Ministry of Skill and development under USTTAD (Upgrading the Skills & Training in Traditional Arts/Crafts for Development) scheme.
- c) It envisages boosting the skills of craftsmen, weavers and artisans who are already engaged in the traditional ancestral work.
- d) None of the above.

Q.71) Who among the following associate with anti-caste revolution against the upper-caste Brahmin communities in Kerala ?

- a) Bal Gangadhar Tilak
- b) V. O. Chidambaram Pillai
- c) Mahatma Gandhi
- d) Sree Narayana Guru

Q.72) Recently developed miSHERLOCK, a small tabletop device that can detect:

- a) Tuberculosis
- b) SARS-CoV-2
- c) HIV- Aids
- d) Malaria

Q.73) Consider the following statements with reference to the Eutelsat Quantum:

- 1. It is the world's first commercial fully re-programmable satellite.
- 2. It has been developed under an European Space Agency partnership project with Airbus.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.74) Consider the following statements with reference to the Sovereign Gold Bond Scheme:

- 1. It is issued by Reserve Bank India on behalf of the Government of India.
- 2. Minimum permissible investment will be 10 gram of gold.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.75) Consider the following statements with reference to the Malabar rebellion:

- 1. It had been an uprising of Muslim tenants against British rulers and local Hindu landlords.
- 2. It has even been described as a Tribal revolt.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.76) Consider the following statements with reference to the Members of Parliament Local Area Development Scheme :

1. Each MP has the choice to suggest to the District Collector for works to the tune of Rs.5 Crores per annum to be taken up in his/her constituency.
2. The Rajya Sabha Members of Parliament may select any one or more Districts from any one State in the Country for implementation of their choice of work under the scheme.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.77) Which of the following statements is incorrect with respect to the Committee on Public Accounts?

- a) It is constituted by Parliament every two year.
- b) It is the oldest Parliamentary Committee and was first constituted in 1921
- c) It scrutinizes the Appropriation Accounts of the Government of India.
- d) It consists of 22 Members, 15 Members are elected by Lok Sabha and 7 Members of the Rajya Sabha

Q.78) Consider the following statements with reference to the whip:

1. A whip is an official of a political party who acts as the party's 'enforcer' inside the legislative assembly or house of parliament.
2. India inherited the concept of the whip from the American Presidential system.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.79) Which of the following is best described as Adjournment?

- a) It means terminating a sitting of Parliament for an indefinite period
- b) It results in the suspension of work in a sitting for a specified time
- c) It means the termination of a session of the House by an order made by the President
- d) It ends the very life of the existing House and a new House is constituted after the General Elections.

Q.80) Which of the following statements is incorrect with respect to the Vehicle scrappage policy ?

- a) The policy aims to scrap old, unfit and polluting vehicles by creating an infrastructure for automated testing of vehicles that have completed the registration period.
- b) The vehicle will then have to undergo a mandatory fitness test.
- c) Owners of such vehicles will get a scrap value which will be equivalent to 4 percent to 6 percent of the ex-showroom price of the new vehicle
- d) There will be one percent registration fees for new vehicles purchased if the owner shows a certificate of deposit

Q.81) Which of the following statements is incorrect with respect to the PM-KUSUM?

- a) The scheme is aimed at ensuring energy security for farmers.
- b) it aims to add solar and other renewable capacity of 25,750 MW by 2024.
- c) State Nodal Agencies (SNAs) of MNRE will coordinate with States/UTs, Discoms and farmers for implementation of the scheme.
- d) None of the above.

Q.82) Consider the following statements with reference to the Zika virus :

1. The Aedes aegypti mosquito is the main vector for spread of disease.
2. It is a cause of microcephaly, congenital nervous system malformations and Guillain-Barré syndrome.

Which of the above statements is/are correct ?



- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.83) Consider the following statements regarding 'Green grid initiative':

- 1 It is an initiative jointly launched by India and France.
- 2 Aim of this initiative is to raise the green fund of 100 billion dollar from the developed countries and spend this in underdeveloped nations for wind and solar energy installment.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.84) Consider the following statements regarding 'Global methane pledge':

- 1 Methane is the most abundant greenhouse gas in the atmosphere.
- 2 It was launched at the COP26 climate conference in Glasgow, Scotland.
- 3 The pledge was first announced by India and the EU.

Which of the statements given above is/are correct?

- a) 1 and 2 Only
- b) 2 Only
- c) 1 and 3 Only
- d) 2 and 3 Only

Q.85) Consider the following statements regarding 'Climate equity monitor':

- 1. It provides an online dashboard for assessing equity in climate action, inequalities in emissions, energy and resource consumption at the National level.
- 2. The website has been conceptualized and developed by M.S Swaminathan Research Foundation (MSSRF), Chennai, and the National Institute of Advanced Studies (NIAS).

Which of the statements given above is/are correct?

- a) 1 Only

- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.86) Recently 'High ambition coalition' was seen in news, consider the following statements:

- 1. The High Ambition Coalition (HAC) for Nature and People is an intergovernmental group of more than 70 countries.
- 2. The central goal of HAC is to protect at least 30 percent of the world's land and ocean by 2030.
- 3. India is the first of the BRICS bloc of major emerging economies to join the HAC.

Which of the statements given above is/are correct?

- a) 1 and 2 Only
- b) 2 and 3 Only
- c) 1 and 3 Only
- d) 1,2 and 3

Q.87) Consider the following statements regarding 'Climate Action and Finance Mobilization Dialogue (CAFMD)':

- 1 It was launched in a joint partnership of India and the United Kingdom.
- 2 The CAFMD is one of the two tracks of the Climate and Clean Energy Agenda 2030 partnership launched at the Leaders' Summit on Climate in April 2021.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.88) Recently COP26 Summit held in Glasgow consider the following statements regarding the India's commitments:

- 1. India will achieve net zero emission by the year 2055.
- 2. India will ensure 50% of its energy will be sourced from renewable sources by the year 2030.
- 3. India will install systems to generate 500 GW of renewable energy by the year 2030.

Which of the statements given above is/are correct?



- a) 1 and 2 Only
- b) 2 and 3 Only
- c) 1 and 3 Only
- d) 1,2 and 3

Q.89) Recently a study pointed out that the area suitable for the survival of the rare sangai deer will become limited in the next few decades, consider the following statements:

- 1. Sangai deer is found only in Manipur, India.
- 2. Sangai is listed as "Endangered" in the International Union for Conservation of Nature Red List of Threatened Species.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.90) Which of the following states has launched the 'CLAP initiative' to boost the cleanliness drive?

- a) Kerala
- b) Andhra Pradesh
- c) Tamil Nadu
- d) Telangana

Q.91) In the context of geodiversity protection consider the following statements:

- 1 India does not have any legislation to protect geological sites.
- 2 India does not have even a single UNESCO designated geopark.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.92) In recent times NGT have taken several steps to safeguard the environment, consider the following statements regarding NGT(National Green Tribunal):

- 1. The National Green Tribunal was established in 2010 under the National Green Tribunal Act.
- 2. National Green Tribunal (NGT) is vested with suo motu power in discharge of its functions.
- 3. The Tribunal is mandated to make an endeavour for disposal of applications or appeals finally within 6 months of filing of the same.

Which of the statements given above is/are correct?

- a) 1 and 2 Only
- b) 2 and 3 Only
- c) 1 and 3 Only
- d) 1,2 and 3

Q.93) With reference to WHO Air Quality Guidelines (AQG), consider the following statements:

- 1) They are applicable to both outdoor and indoor environments globally.
- 2) Unlike all other WHO guidelines, AQGs are legally-binding.
- 3) The move does not have an immediate effect in India as the National Ambient Air Quality Standards do not meet the WHO's existing norms.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2
- c) 1 and 3
- d) 2 and 3

Q.94) SAATH Initiative, which was recently launched, aims at:

- a) It is a Rural Enterprises Acceleration Programme for women associated with Self Help Group (SHGs).
- b) It is an initiative of NITI AYOJ and ROCKY MOUNTAIN INSTITUTE to promote zero-pollution delivery vehicles by working with consumers and industry.
- c) It is a web-based personalized agro-advisory system which uses Information Technology to solve unscientific agricultural practices.
- d) It facilitates officers conduct face-to-face interactions with the Kashmiri youth to express their grievances.



Q.95) It inhabits the Himalayas at elevations ranging from 3,000 to 4,500 m across Jammu and Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh. Listed on appendix 1 of CITES, this is categorized as 'Vulnerable' in IUCN Conservation status. These species are an indicator of the health of high altitude habitats and, increasingly, an important indicator of impacts of climate change on mountain environments. This is also under constant human threat for illegal wildlife trade.

Which of the following species has been described in the above passage?

- a) Northern River Terrapin
- b) Snow Leopard
- c) Kashmir Stag
- d) Manda Buffalo

Q.96) With regard to Renewable Energy Certificates(RECs) , Consider the following statements:

- 1. RECs are traded in power exchange wherein forbearance price and floor price are determined by Central Electricity Regulatory Commission (CERC).
- 2. One Renewable Energy Certificate (REC) is treated as equivalent to 1 GWh.
- 3. The validity of REC is three years
- 4. RECs are regulated by the Ministry of New and Renewable Energy.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 3
- c) 1,2 and 4
- d) 2 and 4

Q.97) Which of the following statement best describe e-RUPI:

- a) It is a cashless and contactless instrument for digital payment
- b) It is a digital currency which is proposed by the RBI
- c) It is a fintech firm with only a digital presence
- d) None of the above

Q.98) In the context of Asia-Pacific Economic Cooperation (APEC), consider the following statements:

- 1. It comprises 21 member states from Asian, American,

and African continents.

- 2. Countries that extend to more than one continent like Russia are not a part of this group.
- 3. Both India and Thailand are not members of this group.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.99) Consider the following pairs with reference to species and their protection status:

Species	Protection Status
1. Saltwater crocodile	— Critically endangered
2. Olive Ridley turtle	— Vulnerable
3. Pygmy Hog	— Endangered

Which of the above pairs is/are correctly matched?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 3 only

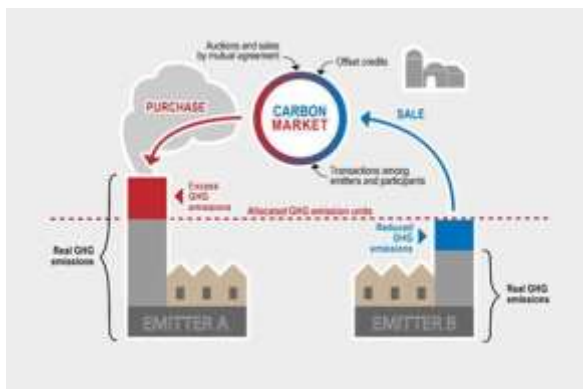
Q.100) Among the following group of countries, which one is a part of the Arctic circle?

- a) USA, Canada, Estonia, Finland
- b) Finland, Denmark, Norway, Sweden
- c) USA, Russia, Belarus, Estonia
- d) Russia, Canada, Estonia, Denmark

Q.1) Ans: c

Exp:

- **Statements 1 and 3 are correct:** Indore-based Enking International may become the world's first company that operates in the carbon markets space to go in for an initial public offering (IPO). Carbon credits are market-tradable instruments like Renewable Energy Certificates that are issued to entities whose projects reduce carbon dioxide emissions.



- **Statement 2 is incorrect:** Carbon trading began under the Kyoto Protocol of 1997 which came into force in 2005. Under this, 'certified emission reductions' or CERs, were issued to entities that put up projects that reduced emissions — such as wind, solar, or energy efficiency.
- It allows nations which are unable to meet their reduction targets to purchase carbon credits.
- The Paris Agreement also allows voluntary trading between countries to meet their NDC goals. If a country reduces more GHG emissions than its target, it can sell the emission reduction to another country as an "internationally traded mitigation outcome".

Extra Edge by Only IAS

In India, Renewable Energy Certificate' (REC) and 'Perform Achieve Trade' (PAT) are energy certification programmes.

Central Electricity Regulatory Commission (CERC) is the Market Regulator for the trading of Energy Savings Certificates (ESCerts).

Q.2) Ans: c

Exp:

- **Both statements 1 and 2 are correct:** NITI Aayog and World Resources Institute (WRI), India, jointly launched the 'Forum for Decarbonizing Transport' in India as part of the NDC-Transport Initiative for Asia (NDC-TIA) project.
- India has a massive and diverse transport sector, which is the third most CO2 emitting sector. This project aims at bringing down the peak level of GHG emissions (transport sector) in Asia.
- It is part of the NDC-Transport Initiative for Asia (NDC-TIA) project. NITI Aayog is the implementing partner for the India component of the project.
- It will help the low-carbon transport system for India. It will facilitate the government with a comprehensive understanding of present as well as future transport activities along with the related CO2 emissions.

Extra Edge by OnlyIAS

The NDC Transport Initiative for Asia (TIA 2020-2023)

It is a joint programme of seven organisations that will engage China, India, and Vietnam in promoting a comprehensive approach to decarbonizing transport in their respective countries. The project is part of the International Climate Initiative (IKI).

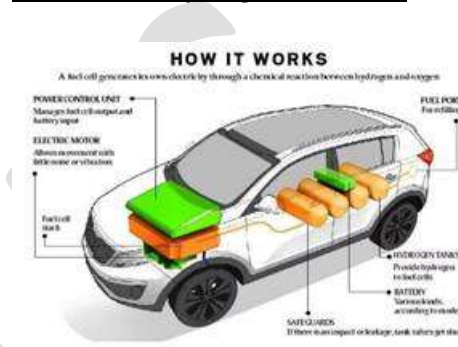
It is a joint project of seven organisations including *World Resources Institute, International Transport Forum, International Council on Clean Transportation etc.*

The NDC-TIA project aims at *promoting a coherent strategy of effective policies for decarbonising transport* that are coordinated among various sector ministries, civil society and the private sector.

Extra Edge By OnlyIAS

Chennai has the **Centre for Fuel Cell Technology, an autonomous R&D centre of the Department of Science and Technology.**

The work of Hydrogen fuel cell :



Hydrogen fuel cells generate electricity through an electrochemical process, **unlike a battery-electricity vehicle, it does not store energy.**

It **relies on a constant supply of fuel and oxygen** — in the same way that an internal combustion engine relies on a constant supply of petrol or diesel, and oxygen.

Unlike the combustion engine cars, **there are no moving parts in the fuel cell**, so they are **more efficient and reliable by comparison.**

Advantages of the Hydrogen fuel cell

Fuel cells have strong advantages over conventional combustion-based technologies

They produce much **smaller quantities of greenhouse gases** and **none of the air pollutants** that cause health problems.

Pure hydrogen is used, **fuel cells emit only heat and water as a byproduct.**

Fuel cells are far **more energy efficient** than traditional combustion technologies.

Q.3) Ans: c

Exp:

- **Statements 1 and 2 are correct:** A Hydrogen fuel cell combines hydrogen and oxygen to generate an electric current, water being the only by-product. At the heart of the fuel cell electric vehicles (FCEV) is a device that uses a source of fuel, such as hydrogen, and an oxidant to create electricity by an electrochemical process. Hydrogen fuel cells convert the chemical energy of fuels directly into electricity. It could come in handy for power generation, transportation using hydrogen as fuel, and in powering electronic instruments like laptops and mobile phones replacing the lithium batteries.
- **Statement 3 is incorrect:** Hydrogen fuel cell is an emerging technology requiring high cost for its applications.

Q.4) Ans: c

Exp:

- The **Prime Minister Narendra Modi** coined the new term “**Surya Putra**” for all the nations which fall between Tropic of Cancer and Tropic of Capricorn.
- The other term used for these countries is “Sunshine Countries”.
- The International Solar Alliance (ISA) was launched by Indian PM Narendra Modi and former French president Francois Hollande in November 2015 at the 21st session of the United Nations Climate Change Conference of the Parties (COP-21) in Paris.
- The primary objective of the alliance is to work for efficient consumption of solar energy to reduce dependence on fossil fuels.
- It comprises solar resource rich countries lying fully or partially between the Tropic of Cancer and Tropic of Capricorn.
- The alliance is dedicated to addressing the special energy needs of ISA member countries. The ISA secretariat has the National Institute of Solar Energy, Gurgaon.
- Objectives of the International Solar Alliance:
 - To force down prices by driving demand
 - To bring standardization in solar technologies to foster research and development.

Extra edge by Only IAS

The alliance was formed with the aim to promote solar energy in 121 member countries and to mobilise over \$1 trillion of investment for the deployment of solar energy at affordable costs.

The alliance is a **treaty-based intergovernmental organization**.

Countries that do not fall within the Tropics can join the alliance and enjoy all benefits as other members, with the exception of voting rights.

After the United Nations, it is the largest grouping of states world-wide.



Global Solar Atlas -

The alliance has partnered with the **World Bank to launch Global Solar Atlas at an ISA event at the World Future Energy Summit in Abu Dhabi.**

Global Solar Atlas is a free online tool that **displays annual average solar power potential at any location in the world** and thus identifies potential sites for solar power generation.

Q.5) Ans: a

Exp:

- **Statements 1 and 2 are correct:** Recently the Union Government launched ‘Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscape’ to reduce emissions from agriculture. The



Green-Ag project seeks to integrate biodiversity, climate change and sustainable land management objectives and practices into Indian agriculture. It aims to bring at least 104,070 ha of farms under sustainable land and water management.

- **Statement 3 is incorrect:** It is funded by the Global Environment Facility (GEF) and the Department of Agriculture, Cooperation, and Farmers' Welfare under the Ministry of Agriculture as the national executing agency. The project would be implemented in collaboration with the Food and Agriculture Organisation (FAO) in high-conservation-value landscapes of five States of following -

(i) Madhya Pradesh: Chambal Landscape

(ii) Mizoram: Dampa Landscape

(iii) Odisha: Similipal Landscape

(iv) Rajasthan: Desert National Park Landscape

v) Uttarakhand: Corbett-Rajaji Landscape.

Extra Edge by Only IAS

India is the **world's third largest** emitter of greenhouse gases (GHGs), after China and the US. The country has pledged a 33-35% reduction in the "emissions intensity" of its economy by 2030, compared to 2005 levels.

Q.6) Ans: d

Exp:

- **National Mission for Sustainable Agriculture (NMSA)** launched in 2014-15. It's aims at making agriculture more productive,

sustainable, remunerative and climate resilient by promoting location specific integrated /composite farming systems; soil and moisture conservation measures; comprehensive soil health management; efficient water management practices.

- It is one of the major missions of the National Action Plan on Climate Change (NAPCC). Change in agricultural practices also plays a crucial role in the mitigation of climate change effects. This mission tries to comprehensively revamp the agricultural practices so that the desired objectives of the Nationally Determined Contributions (NDC's) can be achieved.
- Main Objective of National Mission for Sustainable Agriculture
- To make agriculture more productive, sustainable, remunerative and climate resilient by promoting location specific Integrated/Composite Farming Systems
- To conserve natural resources through appropriate soil and moisture conservation measures
- To adopt comprehensive soil health management practices based on soil fertility maps, soil test based application of macro & micro nutrients, judicious use of fertilizers etc.
- To optimize utilization of water resources through efficient water management to expand coverage for achieving 'more crop per drop'
- To develop capacity of farmers & stakeholders, in conjunction with other on - going Missions e.g. National Mission on Agriculture Extension & Technology, National Food Security Mission, National Initiative for Climate Resilient Agriculture (NICRA) etc., in the domain of climate



Q.7) Ans: d

Exp:

- **Statements 1 is incorrect:** NVVN (NTPC Vidyut Vyapar Nigam), a wholly-owned subsidiary company of NTPC Ltd, in partnership with Energy Efficiency Services Ltd (EESL), launched the Green Charcoal Hackathon.
- **Statements 2 and 3 are correct:** Its aim is to leverage the innovative Indian mind to bridge the technology gap, with the prime objective to clean the air by eliminating farm fire, producing renewable energy out of the agro residue, to promote local entrepreneurship, and to increase the income of the farmers.

Extra Edge by Only IAS

Green Charcoal:

Green Charcoal is a **type of bio-fuel** that can be made locally and inexpensively.

To make this, agricultural waste materials appropriate to the season and the region are carbonized (conversion of an organic substance into carbon or a carbon-containing residue) in a kiln.

It burns cleanly, **reducing exposure to the smoke that causes respiratory infections.**

Source :

<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1677457>

Q.8) Ans: d

Exp:

- **Torrefaction is a thermochemical process** typically at 200-350 °C in the absence of oxygen, at atmospheric pressure with low particle heating rates and a reactor time of one hour.

The process causes biomass to partly decompose, creating torrefied biomass or char, also referred to as 'biocoal'.

- Under torrefaction, it converts biomass into a coal-like material, which has better fuel characteristics than the original biomass.
- Torrefied biomass is more brittle, making grinding easier and less energy intensive.
- Compared to fresh biomass, storage of the torrefied material can be substantially simplified since biological degradation and water uptake is minimized.
- Torrefaction involves the heating of biomass in the absence of oxygen/low oxygen concentrations to a temperature of typically 200 to 400°C.
- During the torrefaction process a combustible gas is released, which is utilised to provide heat to the process.
- Following are benefits of Torrefied Biomass:
 - Higher calorific value
 - More homogeneous product
 - Higher bulk density
 - Excellent grindability
 - Durability
 - Hydrophobic nature/water resistance
 - No biological activity

Q.9) Ans: c

Exp:

- Green crackers don't contain banned chemicals such as lithium, arsenic, barium and lead, leaving a lesser impact on the environment. They are named Safe Water Releaser (SWAS), Safe Thermite Cracker (STAR) and Safe Minimal Aluminium (SAFAL) crackers.



- Green crackers release water vapour and don't allow the dust particles to rise. They are designed to have 30% less particulate matter pollution.
- They have been developed by the National Environmental and Engineering Research Institute (NEERI), a Council of Scientific and Industrial Research lab.
- Green crackers not only reduce air pollution; it has been reported that regular crackers emit about 160 decibels of sound, whereas green crackers have a sound level of 110-125 decibels.
- However, for all their advantages, these eco-friendly firecrackers are more expensive than the regular ones.

Source :

<https://www.firstpost.com/india/what-are-green-crackers-and-what-makes-them-eco-friendly-and-less-polluting-10104851.html/amp>

Q.10) Ans: d

Exp:

Integrated Farming System

- Integrated Farming System (IFS) is an interdependent, interrelated, often interlocking production system based on few crops, animals and related subsidiary enterprises in such a way that maximize the utilization of nutrients of each system and minimize the negative effect of these enterprises on the environment.
- The interrelated, inter-dependent-interlocking nature of Integrated Farming System (IFS) involves the utilization of primary produce and secondary produce of one system, as basic input of the other system, thus making them mutually integrated as one whole unit.

- The agriculture sector contributes 17 per cent of India's total GDP, out of which, 27 per cent comes from animal husbandry and overall, the dairy, poultry and aqua industries contribute 4.4 per cent to the nation's GDP.
- Benefits of Integrated Farming System
- Maximization of yield of all component enterprises to provide steady and stable income at higher levels.
- Rejuvenation/amelioration of systems productivity and achieve agro ecological equilibrium.
- Control the built-up of insects-pest, diseases and weed populations through natural cropping system management and keep them at low level of intensity.
- Reducing the use of chemical fertilizer and other harmful agrochemicals and pesticides to provide pollution free, healthy produce and environment to the society at large.
- Increase in natural resource use efficiency by early recycling of nutrients.
- Mitigation of the negative impact of agriculture or livestock on the environment.
-

Source

<https://www.newindianexpress.com/states/andhra-pradesh/2019/apr/25/integrated-farming-to-double-farmers-income-venkaiah-1968793.amp>

Q.11) Ans: d

Exp:

- **Statement 1 is correct:** The United Nations Convention to Combat Desertification



established in 1994, UNCCD is the sole legally binding international agreement linking environment and development to sustainable and management. UNCCD with 194 Parties is a unique instrument that recognises land degradation as an important factor affecting some of the most vulnerable people and ecosystems in the world.

- **Statement 2 is correct:** The convention aims at adaptation and can, on implementation, significantly contribute to achieving the Millennium Development Goals (MDGs), as well as sustainable development and poverty reduction by means of arresting and reversing land degradation. The UNCCD is particularly committed to a bottom-up approach, encouraging the participation of local people in combating desertification and land degradation.
- **Statement 3 is correct:** The United Nations Convention, The UNCCD is one of the three Rio Conventions, the other two being:
 - **Convention on Biological Diversity (UNCBD)**
 - **United Nations Framework Convention on Climate Change (UNFCCC).**
 - The convention promotes sustainable land management (SLM) as a solution to global challenges.

Source : Environment Book by Shankar IAS
CHAPTER - 28, INTERNATIONAL CONVENTIONS

Q.12. Ans: b

Exp:

TRAFFIC (THE WILDLIFE TRADE MONITORING NETWORK')

TRAFFIC (THE WILDLIFE TRADE MONITORING NETWORK') is a joint conservation programme of WWF and IUCN.

- It aims to ensure that **trade in wild plants and animals is not a threat to the conservation of nature.**
- It was established in 1976 by the Species Survival Commission of IUCN, principally as a response to the entry into force during the previous year of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- TRAFFIC is an **international network, consisting of TRAFFIC International**, based in Cambridge, UK with offices on five continents.
- Since its founding, TRAFFIC has grown to become the **world's largest wildlife trade monitoring programme**, and a global expert on wildlife trade issues.
- This **non-governmental organization** undertakes its activities in close collaboration with governments and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) Secretariat.

Source : Environment Book by Shankar IAS,
CHAPTER - 28, INTERNATIONAL CONVENTIONS

Q.13) Ans: c

Exp:

- **Statement 1 is correct:** The Convention on the Conservation of Migratory Species of Wild Animals (also known as CMS or Bonn Convention) aims to conserve terrestrial, aquatic and avian migratory species throughout their range. It is an intergovernmental treaty,



concluded under the aegis of the United Nations Environment Programme, concerned with the conservation of wildlife and habitats on a global scale. The Convention has membership of 117 Parties from Africa, Central and South America, Asia, Europe and Oceania.

- **Statement 2 is incorrect:** India had become a party to the CMS since November 1, 1983. The Indian Government has signed 'Raptor MoU', on Conservation of Migratory Birds of Prey in Africa and Eurasia, with the Convention on Conservation of Migratory Species (CMS), or Bonn Convention, under the auspices of the United Nations Environment Programme (UNEP).
- **Statement 3 is correct:** The only global convention specializing in the conservation of migratory species, their habitats and migration routes, CMS complements and co-operates with a number of other international organizations, NGOs and partners in the media as well as in the corporate sector.

would significantly benefit from international co-operation are listed in

Appendix II of the Convention.

For this reason, the Convention encourages the Range States to conclude global or regional Agreements.

Source : Environment Book by Shankar IAS,
CHAPTER - 28, INTERNATIONAL CONVENTIONS

Q.14) Ans: c

Exp:

The National Bamboo Mission

- The National Bamboo Mission is a Centrally Sponsored Scheme with 100% contribution from the Central Government.
- It is being implemented by the Horticulture Division under the Department of Agriculture and Cooperation in the Ministry of Agriculture.
- The main objective of the mission is involvement of local people and initiatives for the holistic development of the bamboo sector in terms of growth of bamboo through increase in area coverage.
- Furthermore, it also promotes yields and scientific management, marketing of bamboo and bamboo based handicrafts, generation of employment opportunities etc.
- **Objectives of the Mission :**
 - To **promote the growth of the bamboo sector through an area based regionally differentiated strategy.**
 - To increase the coverage of areas under bamboo in potential areas, with suitable species to enhance yields.

Extra Edge by Only IAS

THE CONVENTION ON THE CONSERVATION OF MIGRATORY SPECIES

Appendix I:

Migratory species **threatened with extinction** are listed on **Appendix I** of the Convention.

CMS Parties strive towards **strictly protecting these animals, conserving or restoring the places** where they live, mitigating obstacles to migration and controlling other factors that might endanger them.

Appendix II:

Migratory species that **need or**



- To promote the marketing of bamboo and bamboo based handicrafts.
- To establish convergence and synergy among stakeholders for the development of bamboo.
- To promote, develop and disseminate technologies through a seamless blend of traditional wisdom and modern scientific knowledge.
- To generate employment opportunities for skilled and unskilled persons, especially unemployed youths.

Source : Environment Book By Shankar IAS
CHAPTER - 26 , INSTITUTIONS AND MEASURES

Q.15) Ans: d

Exp:

- **Pair 1 is correct** : The Stockholm Convention on Persistent Organic Pollutants was adopted at a Conference of Plenipotentiaries on 22 May 2001 in Stockholm, Sweden and entered into force on 17 May 2004. POPs Persistent Organic Pollutants (POPs) are organic chemical substances and they are carbon-based. They possess a particular combination of physical and chemical properties.
 - Remain intact for exceptionally long periods of time (many years)
 - Become widely distributed throughout the environment as a result of natural processes involving soil, water and, most notably, air
 - Accumulate in the fatty tissue of living organisms including humans
 - They are toxic to both humans and wildlife.
- **Pair 2 is incorrect** : **Basel Convention on the Control of Transboundary Movements of Hazardous**

Wastes and their Disposal. This is a comprehensive treaty that was adopted in 1989 by the Conference of Plenipotentiaries. It came into force in 1992. It aims to protect the human environment from the adverse effects of hazardous waste that is generated, managed and disposed of in the world community. It applies the Prior Consent Approval procedure to regulate the transboundary movement of hazardous and other wastes. The Basel Convention **secretariat** is situated in **Geneva, Switzerland**.

- **Pair 3 is correct** : The **Minamata Convention on Mercury** is an international environmental treaty that aims to **protect human health and the environment from the harmful effects of mercury** and its compounds. The Convention was signed in 2013 and entered into force in 2017. It is a UN treaty coming under the United Nations Environment Programme (UNEP). 128 countries are signatories to the Convention, and 119 countries are parties to it. India is a party to the Minamata Convention and ratified it in 2018.

Source : Wikipedia

Q.16) Ans: c

Exp:

Pobitora Wildlife Sanctuary:

- This sanctuary is often referred to as '**Mini Kaziranga**' because its landscape and fauna are similar to **Kaziranga national park**.
 - It harbors the highest density of Rhino in the world and second highest concentration of Rhino in Assam after Kaziranga National Park.
 - Its boundary is made by the river *Garanga Beel on the south and the river Brahmaputra on the North*.
 - Fauna occurred here: Leopard, Asiatic Water buffalo, Jungle Cats, Flying fox, Wild boar, Fishing cat, Short nosed fruit bat, Barking deer, Grey mask



shrew etc.

Source :

<https://www.thehindu.com/news/national/other-states/mini-kaziranga-has-58-waterfowl-species/article33786976.ece/amp/>

Q.17) Ans: b

Exp:

- **Statement 1 is incorrect** : Recently, **Kasarkod and Padubidri beaches in Karnataka** were accorded '**Blue Flag**' tag from the international agency **Foundation for Environment Education (FEE), Denmark**. The 'Blue Flag' is a certification that can be obtained by a *beach, marina, or sustainable boating tourism operator*, and serves as an eco-label.
- The certification is awarded annually by the Denmark-based non-profit Foundation for Environmental Education (FEE).

Q.18) Ans: c

Exp:

- **PARIVESH** (Proactive and Responsive facilitation by Interactive, Virtuous and Environmental Single-window Hub) is a **Single-Window Integrate Environmental Management System**.
- PARIVESH is a web-based, role-based workflow application.
- It was launched on the occasion of World Biofuel Day i.e. 10 August 2018 and is developed by the Ministry of Environment, Forest and Climate Change, along with the technical assistance of the National Informatics Centre, (NIC).
- Single registration and single sign-in for all types of clearances (i.e. Environment, Forest, Wildlife and CRZ), unique-ID for all types of clearances required for a particular project.
- A single Window interface for the proponent to submit applications for getting all types of clearance (i.e. Environment, Forests, Wildlife and CRZ clearan

ces).

Source : Environment Book by Shankar IAS
CHAPTER - 8 ENVIRONMENTAL IMPACT ASSESSMENT

Q.19) Ans: c

Exp:

- **India Water Resources Information System** is a **web portal containing information related to Water Resources** through dashboards for rainfall, water levels & discharge of rivers, water bodies, ground water levels, reservoir storages, evapotranspiration and soil moisture, as well as modules on water resources projects, water bodies, hydro-met data availability and tools for GIS layer editing.
- **The Ministry of Jal Shakti** has launched a new version of '**India Water Resources Information System**' (**India-WRIS**). India-WRIS was launched in 2019 under the National Hydrology Project.
- The Ministry of Jal Shakti has established a dedicated organization, **National Water Informatics Centre (NWIC) to maintain and update India's WRIS**. The basis for all hydro-met observation data is the Water Information Management System (WIMS).
- The Water Information Management System (WIMS) receives data from many central and state agencies like CWC, CGWB, IMD, NRSC, Andhra Pradesh, Uttar Pradesh and Gujarat etc.

Source

: <https://pib.gov.in/PressReleasePage.aspx?PRID=1643776>

Q.20) Ans: d

Exp:

Paramagnetism

- **Paramagnetism is derived** from atoms, molecules, or ions possessing a permanent magnetic moment associated with unpaired electron spins.
- **Paramagnetic polarization consists of the**

alignment of such magnetic dipoles which have random orientations under normal conditions.

- Materials that display paramagnetism are called "paramagnetic".
- Examples of paramagnetic coordination complex myoglobin, transition metal complexes, iron oxide (FeO), and oxygen (O₂). **Titanium and aluminum are metallic elements that are paramagnetic.**
- Paramagnetic materials include most chemical elements and some compounds, they have a relative magnetic permeability slightly greater than 1, hence are attracted to magnetic fields.
- **Works of Paramagnetism**
- Paramagnetism results from the presence of at least one unpaired electron spin in a material's atoms or molecules. Any material that possesses atoms with incompletely filled atomic orbitals is paramagnetic.
- The spin of the unpaired electrons gives them a magnetic dipole moment. "Basically, each unpaired electron acts as a tiny magnet within the material. When an external magnetic field is applied, the spin of the electrons aligns with the field. Because all the unpaired electrons align the same way, the material is attracted to the field.
- When the external field is removed, the spins return to their randomized orientations.

Q.21) Ans: c

Exp:

Zero Liquid Discharge (ZLD)

- **Zero Liquid Discharge (ZLD)** is a treatment process designed to **remove all the liquid waste from a system**. The focus of ZLD is to reduce wastewater economically and produce clean water that is suitable for reuse (e.g. irrigation).
- The advantages of it are *saving money and being beneficial to the environment*.
- ZLD systems employ **advanced wastewater/desalination treatment technologies to purify and recycle virtually** all of the wastewater produced.
- *ZLD technologies help plants meet discharge and*

water reuse requirements, enabling businesses to:

- o Meet stringent government discharge regulations
- o Reach higher water recovery (%)
- o Treat and recover valuable materials from the wastewater streams, such as potassium sulfate, caustic soda, sodium sulfate, lithium and gypsum.
- The conventional way to reach ZLD is with thermal technologies such as **evaporators** (multi stage flash (MSF), **multi effect distillation** (MED) and **mechanical vapor compression** (MCV)) and **crystallizers**, And then ZLD plants produce solid waste.

Source : Wikipedia

Q.22) Ans: c

Exp:

- It is **India's only Golden Tiger** found in **Kaziranga National Park of Assam**.

A golden tiger also known as tabby tiger or strawberry tiger, Golden colour tiger caused by a recessive gene.

- The yellow skin of tigers is controlled by a set of 'agouti genes' while the black stripes are controlled by 'tabby genes' and their alleles.

- Suppression of any of these genes may lead to color variation in a tiger
- Golden tigers are characterized by *blonde or pale-golden color and reddish stripes* in place of black like in normal tigers.



Source : <https://indianexpress.com/article/trending/trending-in-india/look-at-this-beauty-picture-of-golden-tiger-in-assam-goes-viral-6502244/lite/>

Q.23) Ans: c

Exp:

- **India's first snow leopard conservation centre will be established in Uttarakhand.** It was started in 2017.



- The conservation centre will be built by the Uttarakhand forest department along United Nations Development Programme (UNDP) in Uttarkashi forests as part of its six-year long project, SECURE Himalayas.

- The project looks into conservation of snow leopards and other endangered species and their habitats, found in the Himalayas.

- It is funded by the Global Environment Facility-United Nations Development Programme.

- **Snow Leopards :**

- o It is a Schedule I animal under the Wildlife Protection Act of India.

- o IUCN status: Vulnerable

- o They are listed under Appendix I of CITES.

- **Conservation Efforts by India:**

- o The Government of India has identified the snow leopard as a flagship species for the high altitude Himalayas.

- India has also been a party to the Global Snow Leopard and Ecosystem Protection (GSLEP) Programme since 2013.

Q.24) Ans: d

Exp:

- **Joint Forest Management (JFM)** : It is a **forest movement** involving both the **state forest departments and local communities**. The policies and objectives of the Joint Forest Movement are detailed in the Indian comprehensive National Forest Policy of 1988 and the Joint Forest Management Guidelines of 1990 of the Government of India.

- Villagers agree to assist in the safeguarding of forest resources through protection from fire, grazing, and illegal harvesting in exchange for which they receive non-timber forest products and a share of the revenue from the sale of timber products.

- JFM is a landmark initiative in the history of Indian forestry. However, its impact remains limited due to the following reason:

1. Lack of support

2. **Bureaucratic Centralization**

3. **Conflict with Panchayats**

4. Poor monitoring and lack of convergence

5. Inequitable benefit sharing

6. Unequal partnership loaded in favour of officials.

Source : Wikipedia

Q.25) Ans: b

Exp:

- **Statement 1 is incorrect-** State of environment survey is an annual publication by the Centre for Science and Environment(CSE) along with Down To Earth. It covers aspects such as forests, wildlife, agriculture, rural development, water and sanitation, and climate change. This publication is regarded as the most credible and complete annual survey of India's environment.

- **Statement 2 is correct-** India's air, water and land have become more polluted between 2009 and 2018. Tarapur in Maharashtra emerged as the most polluted cluster.

Performance of various states -

- o When ranked on the basis of achieving Sustainable Development Goals: The best performing States were Kerala, Himachal Pradesh, Andhra Pradesh, Tamil Nadu and Telangana. The worst performers were Bihar, Jharkhand, Arunachal Pradesh, Meghalaya and Uttar Pradesh.

Extra edge by Only IAS

Centre for Science and Environment (CSE)

- It is a Non-Profit organisation.
- It is Based in New Delhi.
- It was Established in 1980.
- CSE works as a think tank on environment-development issues in India.

Reference: <https://www.newindianexpress.com/nation/2021/feb/26/pandemic-will-leave-375-million-children-with-health-impactsstate-of-indias-environment-2021-report-2269142.html>



Q.26) Ans: c

Exp:

- **Statement 1 is incorrect** : Koala is an Australian animal, often called the koala bear. They are marsupials (mammals with a pouch for the development of offspring).

- **Statement 2 is correct** : They are marsupials (mammals with a pouch for the development of offspring). Their fur (hair) is more like the coarse wool of a sheep. They have two toes that are fused together on their feet, which they use to comb their fur.

- **Statement 3 is correct** : Eucalyptus forests of southeastern and eastern Australia. Koalas rely on the eucalyptus tree for both habitat and food. IUCN Red List Status - **Vulnerable**.

Threats-

- o Hunting for their fur.
- o Habitat loss due to land clearing, logging, and bushfires.
- o Threats of predation and road accidents.

Reference: <https://www.britannica.com/animal/koala>

Q.27) Ans: b

Exp:

- **Statement 1 is correct**: Sea snout is a slimy layer of grey or green sludge, which can cause considerable damage to the marine ecosystem. It is formed **when algae are overloaded with nutrients**.

- **Statement 2 is incorrect**: A 'sea snout' outbreak was first recorded in the country in 2007. Back then, it was also spotted in the Aegean Sea near Greece. Overloading of nutrients happens because of warm weather caused by global warming, water pollution, uncontrolled dumping of household and industrial waste into the seas etc.

- **Statement 3 is correct**: The growth of the **mucilage**, which floats upon the surface of the sea like brown phlegm, is posing a severe threat to the marine ecosystem of the country. It has caused **mass deaths among the fish population, and also killed other aquatic organisms such as corals and sponges**. The mucilage is now covering the surface of the sea and has also spread to 80-100 feet below the

surface. If unchecked, this can collapse to the bottom and cover the sea floor, causing major damage to the marine ecosystem. Over a period of time, it could end up poisoning all aquatic life, including fishes, crabs, oysters, mussels and sea stars.

Reference: <https://indianexpress.com/article/explained/explained-what-is-the-sea-snot-outbreak-in-turkey-7347989/>

Q.28) Ans: d

Exp:

- **Statement 1 is correct** : Ambergris, French for grey amber, is generally referred to as **whale vomit**. It is a solid **waxy substance that floats around the surface of the water body** and at times settles on the coast. A sperm whale eats several thousand squid beaks a day. Occasionally, a beak makes its way to the whale's stomach and into its looping convoluted intestines where it becomes ambergris through a complex process, and may ultimately be excreted by the whale.

- **Statement 2 is correct** : It floats around the surface of the water body and at times settles on the coast. Because of its high value it is referred to as **floating gold**. **1 kg of ambergris is worth Rs 1 crore** in the international market.

- **Statement 3 is correct** : It is used in the **perfume market**, especially to create fragrances like musk. It is believed to be in high demand in countries like Dubai that have a large perfume market. Ancient Egyptians used it as **incense**. It is also believed to be used in some **traditional medicines and as a spice**.

Reference: <https://www.britannica.com/science/ambergris>

Q.29) Ans: d

Exp:

- A Positron is an **antimatter** counterpart to an electron. A positron has the equal or same mass as an electron. But the electron has a Negative Electric Charge whereas Positron has a Positive electric Charge.

- **Positron Excess**- Over the years scientists observed



excess positrons having the energy of more than 10 GeV. But Positrons having an energy of more than 300 GeV are lower in comparison to astronomers' expectations. This behavior of positrons **between 10 and 300 GeV is called the 'positron excess'.**

- **Reason for Positron Excess:** The Milky Way consists of giant clouds of molecular hydrogen. They are the seats of the formation of new stars. These clouds can be as massive as 10 million times of the Sun's mass. Further, these clouds can extend to 600 light-years. The Milky Way is a huge collection of stars, dust, and gas. It is called a spiral galaxy because if viewed from the top or bottom, it will look like a spinning pinwheel. Cosmic rays are produced in supernovae explosions. These waves propagate through these giant clouds before they reach the Earth.

- Cosmic rays interact with **molecular hydrogen** and can give rise to other cosmic rays primarily electrons and positrons. Further, cosmic rays propagate through these clouds and decay from their original forms and lose their wave energy by energising the clouds. Some may also get re-energised. Researchers from Raman research institute found out that Cosmic rays interaction with giant clouds is a viable contributor to the origin of **positron excess phenomenon.**

Reference: <https://pib.gov.in/PressReleasePage.aspx?PRID=1715111>

Q.30) Ans: d

Exp:

- The International system of units (SI), commonly known as the **metric system**, is the international standard of measurement. The International treaty of the meter was signed in **Paris on May 20, 1875** by seventeen countries, including the United States and is now celebrated around the globe as world metrology day.
- The SI rests on a foundation of seven defining constants: the caesium hyperfine splitting frequency, the speed of light in vacuum, the Planck constant, the elementary charge (i.e. the charge on a proton) the Boltzmann constant, the Avogadro constant, and the

luminous efficacy of a specified monochromatic source.

- Definitions of all seven SI base units are expressed using an explicit-constant formulation and experimental realized using specific practical techniques.
- The seven SI base units, which are comprised of:
 - o **Length- meter (m)**
 - o Time- second (s)
 - o Amount of substance- mole
 - o **Electric current- ampere (A)**
 - o Luminous intensity- candela (cd)
 - o **Mass- kilograms (kg)**
- Hence only 1,3 and 4 options are correct.

Reference: <https://www.nist.gov/pml/weights-and-measures/metric-si/si-units>

Q.31) Ans: a

Exp:

- **Statement 1 is correct :** Ebola virus disease First discovered in 1976 near the **Ebola River** in what is now the **Democratic Republic of Congo**. Ebola virus disease (EVD), formerly known as Ebola haemorrhagic fever, is a severe, often fatal illness in humans.
- **Statement 2 is correct :** The virus is transmitted to people from **wild animals and spreads in the human population through human-to-human transmission.** The average EVD case fatality rate is around 50%. Case fatality rates have varied from 25% to 90% in past outbreaks. Early supportive care with rehydration, symptomatic treatment improves survival. There is yet no licensed treatment proven to neutralise the virus but a range of blood, immunological and drug therapies are under development.
- **Statement 3 is incorrect :** An experimental **Ebola vaccine**, called rVSV-ZEBOV proved highly protective against EVD in a major trial in Guinea in 2015. The rVSV-ZEBOV vaccine is being used in the ongoing 2018-2019 Ebola outbreak in DRC. Pregnant and breastfeeding women should have access to the vaccine under the same conditions as for the general population.

Reference: <https://www.who.int/news-room/fact-sheets/detail/ebola-virus-disease>



Q.32) Ans: b

Exp:

- **Pusa bio-decomposer** is a solution developed by the scientists at Indian Agricultural Research Institute, Pusa, which can turn crop residue into manure in 15 to 20 days and therefore, can prevent stubble burning.
- When mixed in a water solution and sprayed on land, it softens and decomposes the paddy subtly to the extent that it can mix with soil and act as compost. Theoretically, it does away with the reason farmers set fire to fields ahead of the rabi sowing. It is essentially a fungi-based liquid solution that can soften hard stubble to the extent that it can be easily mixed with soil in the field to act as compost.
- This would then rule out the need to burn the stubble, and also help in retaining the essential microbes and nutrients in soil that are otherwise damaged when the residue is burned. These seven strains of fungi are packed into four capsules, which cost about Rs 20 per pack of four.

Reference: <https://www.thehindu.com/news/cities/Delhi/stubble-burning-pusa-bio-decomposer-successful-will-tell-supreme-court-says-kejriwal/article33022155.ece>

Q.33) Ans: a

Exp:

- **Statement 1 is correct** : Protons, neutrons, and electrons are tiny particles that make up atoms. The neutrino is also a tiny elementary particle, but it is not part of the atom. Neutrino has a **very tiny mass, no charge and spins half**. It interacts very weakly with other matter particles.
- **Statement 2 is incorrect** : Neutrinos come from the sun (solar neutrinos) and other stars, cosmic rays that come from beyond the solar system, and from the Big Bang from which our Universe originated. They can **also be produced in the lab**.
- Neutrinos come in three '**flavours**', another name for 'types', and each flavour is associated with a light elementary particle. For instance, the electron-neutrino is associated with the electron; the muon-neutrino with the muon and the tau-neutrino with the tau particle.

Reference: <https://www.scientificamerican.com/article/what-is-a-neutrino/>

Q.34) Ans: a

Exp:

About Hope Mars mission :

- The Hope mission is an unmanned space probe named **Al-Amal**, meaning "Hope" in Arabic. The Hope mission is a Mars orbiter spacecraft, which will study the thin atmosphere of Mars. It is the first interplanetary mission for the Arab World.
- The three main objectives of the Hope probe are -
 - o **Statement 1 is correct-** To understand the **climate dynamics and global weather** map of Mars by studying the lower atmosphere of Mars.
 - o **Statement 2 is incorrect-** To understand the presence and variability of **hydrogen and oxygen in the upper atmosphere, and why Mars is losing these gases to space**.
 - o **Statement 3 is correct-** To explain how the weather of Mars affects the escape of hydrogen and oxygen, by correlating conditions in the lower and upper atmosphere
- It is a known fact that the Red Planet was once habitable, from signatures of flowing water and organic material that point to a past that could have supported living things. An understanding of Mars' past could help scientists understand the future of Earth.

Reference: <https://www.space.com/hope-mars-mission-uae>

Q.35) Ans: d

Exp:

- **Russia** has successfully launched the Arktika-M satellite. The satellite was launched from the **Soyuz carrier rocket** from the **Baikonur Cosmodrome in Kazakhstan**.
- Arktika-M satellite is a **remote-sensing and emergency communications** satellite. It is designed to monitor the **climate and environment in the Arctic**



region. Russia plans to send up a second satellite in 2023. This satellite will be combined with the Arktika-M Satellite.

- These two satellites will offer round-the-clock, all-weather monitoring of the Arctic Ocean and the surface of the Earth. The satellite will help create more accurate models for compiling short-term weather forecasts. The compilation will give researchers a large amount of new data for studying global climate change.

Reference: <https://www.thehindu.com/sci-tech/science/russia-launches-satellite-to-monitor-climate-in-arctic/article33959352.ece>

Q.36) Ans: a

Exp:

- **Statement 1 is correct :** They are thought to be a result of the **solar wind and Earth's magnetic field interacting**. It is a huge, funnel-like, spiral geomagnetic storm that occurs above the polar ionosphere of Earth, during extremely quiet conditions.
- **Statement 2 is correct :** They are made up of **plasmas, consisting of extremely hot ionized gases** that rotate at extremely high speeds. Space hurricanes are caused by plasma unleashed from the sun as solar wind. These charged particle clouds travel through space and fuel magnetic storms as they interact with magnetic fields.
- **Statement 3 is incorrect :** They are related to the **aurora borealis** phenomenon, as the electron precipitation from the storm's funnel produces gigantic, cyclone-shaped auroras.
- The researchers think these kinds of storms could create more drag on satellites and interfere with radio signals and communications, making these events particularly important to understand.

Reference: <https://physicsworld.com/a/space-hurricane-observed-in-the-earths-upper-atmosphere/>

Q.37) Ans: b

Exp:

- **Statement 1 is incorrect :** Coronaviruses are a

group of related RNA viruses that cause diseases in **mammals and birds**. In humans and birds, they cause respiratory tract infections that can range from mild to lethal.

- **Statement 2 is correct :** Experts believe the virus that causes COVID-19 spreads mainly from **person to person**. There are several ways this can happen:
 - o **Droplets or aerosols-** This is the most common transmission. When an infected person **coughs, sneezes, or talks, droplets or tiny particles called aerosols** carry the virus into the air from their nose or mouth. Anyone who is within 6 feet of that person can breathe it into their lungs.
 - o **Airborne transmission-** Research shows that the virus can live in the air for up to 3 hours.
- **Statement 3 is correct :** HIV is a **retrovirus**, which means it carries **single-stranded RNA as its genetic material** rather than the double-stranded DNA human cells carry. Retroviruses also have the enzyme reverse transcriptase, which allows it to copy RNA into DNA and use that DNA "copy" to infect human, or host, cells. When HIV infects a cell, it first attaches to and fuses with the host cell.
- **Statement 4 is incorrect :** Rabies is a preventable **viral disease** most often transmitted through the bite of a rabid animal. The rabies virus infects the **central nervous system of mammals**, ultimately causing disease in the brain and death. The vast majority of rabies cases reported to the Centers for Disease Control and Prevention (CDC) each year occur in wild animals like **bats, raccoons, skunks, and foxes**, although any mammal can get rabies.

Q.38) Ans: a

Exp:

- **Statement 1 is correct-** A cruise missile is a guided missile (target has to be pre-set) used against **terrestrial targets**. It remains in the atmosphere throughout its flight. Cruise missile flies within the earth's atmosphere and uses jet engine technology. Cruise missiles are known specifically for low-level flight which is staying relatively close to the surface of the earth to avoid detection from anti missile systems.



- **Statement 2 is incorrect-** Cruise Missile is a guided missile that flies with constant speed to deliver a warhead at specified target over **long distance with high accuracy**. Cruise missiles are designed to deliver a **large warhead** over long distances with high precision. Modern cruise missiles are capable of travelling at **supersonic or high subsonic speeds**, are self-navigating, and are able to fly on a non-ballistic, extremely low-altitude trajectory.

Reference: <https://www.britannica.com/technology/cruise-missile>

Q.39) Ans: a

Exp:

About Baikal-GVD :

- Baikal-GVD is a **giant underwater space telescope in Lake Baikal**, developed by Russian scientists. It will search for neutrinos, which are nearly **massless subatomic particles** with no electrical charge. Clear freshwater and thick, protective ice cover make Lake Baikal an ideal place to search for neutrinos.
- **Neutrinos-** Neutrinos are the smallest particles currently known. Neutrinos are everywhere, but they interact so weakly with the forces around hence, are hard to detect.

Extra edge by Onlyias

Lake Baikal-

- Situated in **south-east Siberia**, the 3.15-million-ha lake is the oldest (25 million years) and deepest (1,700 m) lake in the world.
- It contains 20% of the world's total unfrozen freshwater reserve.
- Known as the '**Galapagos of Russia**', its age and isolation have produced one of the world's richest and most unusual freshwater faunas. It is home to approximately 1,700 to 1,800 endemic plant and animal species.
- In 1996, it was declared a **UNESCO World Heritage Site**.

Reference: <https://baikalgvd.jinr.ru/>

Q.40) Ans: c

Exp:

- **Statement 1 is correct** : Human in space policy for India guidelines are applicable to activities presently postulated under Indian human space program (IHSP) and for those activities which shall be defined and carried out in future in conjunction with IHSP.
- **Statement 2 is correct** : the aim of the policy envisages undertaking demonstration of human spaceflight to LEO (low-earth orbit) in the short-term and laying the foundation for a sustained exploration in the long run that will enable presence in LEO and beyond.

Extra edge by Only IAS

Indian Human Spaceflight Programme (IHSP):

- IHSP was initiated in 2007 by the **Indian Space Research Organisation (ISRO)** to develop the technology needed to launch crewed orbital spacecraft into Low Earth orbit.
- Under it, the first unmanned mission is planned for December 2021. Second **unmanned flight is planned in 2022-23**, followed by human spaceflight demonstration (Gaganyaan).
- **Gaganyaan** has been designed to carry three Indian astronauts to low earth orbit (an orbit of 300-400 km) for a period of five to seven days.
- Gaganyaan National Advisory Council has been created with members from different institutions and industries to deliberate on various aspects of the mission.
- Gaganyaan includes a Space Capsule Recovery Experiment (SRE- 2007), Crew module Atmospheric Re entry Experiment (CARE-2014), GSLV Mk-III (2014), Reusable Launch Vehicle Technology Demonstrator (RLV-TD), Crew Escape System and Pad Abort.

Reference: <https://www.livemint.com/science/news/gaganyaan-is-just-the-beginning-india-plans-to-have-sustained-human-presence-in-space->



[11613380982659.html](https://www.onlyias.com/11613380982659.html)

Q.41) Ans: b

Exp:

- **Statement 1 is incorrect**- Three scientists - **Syukuro Manabe and Klaus Hasselmann and Giorgio Parisi** - have been awarded the Nobel Prize for physics 2021.
 - o Manabe and Hasselmann were awarded for their work in "the physical modelling of Earth's climate, quantifying variability and reliably predicting global warming".
 - o Parisi was awarded for "the discovery of the interplay of disorder and fluctuations in physical systems from atomic to planetary scales."
- **Statement 2 is correct** - This is the first time climate scientists have been awarded the Physics Nobel. Manabe and Hasselmann demonstrated how increases in the amount of carbon dioxide in the atmosphere would increase global temperatures, laying the foundations for current climate models.

Extra edge by Onlyias

Climate Science and Nobel Prize Recognition:

- The **Intergovernmental Panel on Climate Change (IPCC)** had won the Peace Nobel in 2007, an acknowledgement of its efforts in creating awareness for the fight against climate change.
- A **Chemistry Nobel to Paul Crutzen** in 1995, for his work on the ozone layer, is considered the only other time someone from atmospheric sciences has won this honour.

Reference: <https://www.nobelprize.org/prizes/lists/all-nobel-prizes-in-physics/>

Q.42) Ans: c

Exp:

About SC120-LOX

- Hindustan Aeronautics Ltd (HAL) has delivered the 'heaviest' Semi-Cryogenic propellant tank (SC120-LOX) to the Indian Space Research Organisation (ISRO). The **semi cryo-liquid oxygen (LOX) tank** - the first developmental welded hardware - is a part of the SC120 stage.

- **SC120 stage** is intended for payload enhancement by replacing the L110 stage in the existing GSLV Mk-III launch vehicle.
 - **Hindustan Aeronautics Limited, is a strategic** reliable partner to ISRO, has delivered critical structures, tankages, satellite structures for the PSLV, GSLV-Mk II and GSLV-Mk III launch vehicles since the last 5 decades.
 - **Semi Cryogenic Engine**- Unlike a Cryogenic engine, a Semi Cryogenic engine uses Refined kerosene instead of liquid hydrogen. The liquid oxygen is used as a **Oxidiser**. That's the advantage of using a Semi Cryogenic engine as it requires Refined Kerosene which is lighter than liquid fuel and can be stored in a normal temperature.
- Reference:** <https://m.economictimes.com/news/science/hal-delivers-heaviest-semi-cryogenic-propellant-tank-to-isro/articleshow/86838143.cms>

Q.43) Ans: c

Exp:

- **About Epilepsy**- Epilepsy is a central nervous system (**neurological**) **disorder** in which brain activity becomes abnormal, causing seizures or periods of unusual behavior, sensations, and sometimes loss of awareness. Epilepsy is the fourth most common neurological disorder and affects people of all ages.
- **Symptoms**- Alterations to sense of taste, smell, sight, hearing, or touch, dizziness, tingling and twitching of limbs, Staring blankly, unresponsiveness, performing repetitive movements. It may or may not involve loss of awareness or consciousness.
- There's no cure for epilepsy, but the disorder can be managed with medications and other strategies. In 2019, a report Epilepsy, a public health imperative was released by the **World Health Organization**- It is the first global report on epilepsy summarizing the available evidence on the burden of epilepsy and the public health response required at global, regional and national levels.

Reference: <https://www.epilepsy.com/learn/about-epilepsy-basics/what-epilepsy>



Q.44) Ans: d

Exp:

· **Statement 1 is Incorrect:** Scientists have used the term “**hybrid immunity**” to refer to the immune-strengthening effect of exposure to **infection followed by vaccination**. They have drawn this metaphor from genetics. In plants, for example, when the offspring of two breeds grow more strongly than either parent, this is known as “**hybrid vigor**”. Research suggests that people who have had COVID-19 and later receive the first dose of an mRNA vaccine develop exceptional immunity against a wide range of variants of the virus.

· **Statement 2 is Incorrect:** The immunological advantage from hybrid Immunity arises mostly from memory B cells which evolve in the lymph nodes. The bulk of antibodies after infection or vaccination decline after a short while; however, the **memory B cells get triggered** on Subsequent infection or vaccination.

Extra edge by Onlyias

About Memory cell:

· A long-lived immune cell that has the ability to recognize a foreign particle that it previously encountered and consequently mount a faster and stronger immune response.

· These immune cells do not respond immediately when it first encounters an antigen but facilitates a more rapid secondary response when the antigen is encountered on a subsequent occasion. Examples of memory cells are memory B cells and memory T cells.

· **Memory B cells:** They are clones of a parent B cell that previously served as an antigen-presenting cell and then activated by a helper T cell to proliferate. As clones, the memory B cells bear the same B cell receptors as those of the parent B cell. Therefore, they would be able to detect the same antigen when re-exposed. Memory B cells produce more robust antibody-mediated immune responses during re-infection.

· **Memory T cells:** They are T lymphocytes that have the similar capability of recognizing foreign particles that they previously encountered. But unlike memory B cells, the memory T cells do not produce antibodies. Re-exposure to the pathogens causes them to clone

themselves immediately and as such respond to the infection more strongly.

Reference:

<https://www.medicalnewstoday.com/articles/vaccination-gives-hybrid-immunity-to-recovered-covid-19-patients#Antibody-stockpiles>

<https://www.biologyonline.com/dictionary/memory-cell>

Q.45) Ans: c

Exp:

· **Statement 1 is Correct:** Zeolites are crystalline solids structures made of **silicon, aluminum and oxygen** that form a framework with cavities and channels inside where cations, water and/or small molecules may reside. They are often also referred to as molecular sieves.

· **Statement 2 is correct:** Many of them occur naturally as minerals, and are extensively mined in many parts of the world finding applications in industry and medicine. However, most zeolites have been made synthetically, some of them made for commercial use while others created by scientists to study their chemistry. At present, **there are 191 unique zeolite frameworks identified, and over 40 naturally occurring zeolite frameworks** are known.

Extra edge by Onlyias

About Zeolites: Zeolites were introduced in 1954 as adsorbents for industrial separations and purifications. Because of their unique porous properties. Zeolites are used now in a variety of applications with world production estimated to be in the range of over 2.5 million to 3 million metric tons (Mt).

Applications of zeolites:

- They are used in petrochemical cracking.
- Water softening and purification.
- In the separation and removal of gases and



solvents.

- Agriculture, animal husbandry and construction.
- **Use in oxygen concentrator:** Zeolite beads separate oxygen from nitrogen in air by tightly binding to hydrogen, while Oxygen remains free.

Reference:

<https://asdn.net/asdn/chemistry/zeolites.php>

<https://cosmic-etn.eu/%EF%BB%BFzeolites-for-health-care-oxygen-concentrator/>

Q.46) Ans: c

Exp:

- **Statement 1 is correct:** Nano Urea (liquid) is a nano fertilizer. It contains nano nitrogen particles of size range (20-50 nm) dispersed in water. Total nitrogen concentration in one bottle of Nano Urea (liquid) is 4 % (40,000 ppm).

- **Statement 2 is correct:** IFFCO developed nanotechnology based Nano Urea (Liquid) fertiliser to address the imbalance and excessive use of conventional Urea. This nanofertilizers has been developed **indigenously**, for the **first time in the world** at IFFCO - Nano Biotechnology Research Centre (NBRC) Kalol, Gujarat through a proprietary patented technology.

- **Statement 3 is Incorrect:** According to a release from IFFCO, field trials have shown that a 500 ml bottle of nano urea can replace one bag of conventional urea as it has 40,000 ppm of nitrogen, which is equivalent nitrogen nutrient provided by one bag of conventional urea. And it is more favourable over conventional urea as its **500 ml bottle will be priced at Rs 240** which is **10% less** than a bag of conventional urea.

Extra edge by Onlyias

Benefits of nano urea:

Nano Urea (liquid) when sprayed on the leaves easily enters through stomata and other openings and fulfils nitrogen requirement of crops. Because of its unique size and surface area to volume ratio, it effectively addresses crop nutrient requirements. It results in reduced nutritional stress, better growth and yield attributes of crops.

In field trials, it was also found that nano urea is further set to replace conventional urea as it can curtail the requirement of the same by at least 50 per cent.

A practical field trial of **Drone Spraying** of Nano Liquid Urea was conducted in Bhavnagar, Gujarat. The trial was attended by a large number of farmers. This demonstration of spraying liquid nano urea by drone was undertaken by IFFCO.

Reference:

<https://nanourea.in/>

<https://theprint.in/india/what-is-nano-urea-indias-21st-century-product-aiming-to-revolutionise-world-agriculture/673151/>

<https://pib.gov.in/PressReleasePage.aspx?PRID=1760014>

Q.47) Ans: c

Exp:

- **Statement 1 is correct:** 'Main Bhi Digital 3.0' - A special Campaign for Digital Onboarding and Training for street vendors under **PM SVANidhi scheme**. **BharatPe, Mswipe, PhonePe, Paytm, Aceware** are participating in this drive to issue UPI IDs, QR code and provide digital training. Digital Payment Aggregators will handhold the street vendors to bring about enhanced adoption of digital transactions and behaviour change.

- **Statement 2 is correct:** **Ministry of Housing and Urban Affairs(MoHUA)** in collaboration with the **Ministry of Electronics & Information Technology (MeitY)** launched the pilot drive 'Main Bhi Digital 3.0'.



- **Statement 3 is Incorrect:** This campaign has been launched under **PM SVANidhi scheme** across 223 cities in the country.

Extra edge by Onlyias

About PM SVANidhi scheme:

- PM Street Vendor's AtmaNirbhar Nidhi (**PM SVANidhi**) was launched on **June 1, 2020** as a Central Sector Scheme.
- The scheme facilitates affordable **working capital loans of up to ₹10,000 with an interest subsidy @7% on regular repayment.**
- The street vendors are not required to pay any collateral for the loan.
- The scheme provides Graded Guarantee Cover, to Lending Institutions, on a portfolio basis.
- The scheme targets to cover **50 lakh street vendors**, who had been vending on or before March 24, 2020.
- The street vendors also stand a chance to avail a **loan of ₹20,000 and ₹50,000 in the second and third tranches** respectively on timely repayment of the loan.
- To encourage digital transactions a **cash back up to ₹1,200 on digital transactions at ₹100 per month** is provided to the street vendors under the scheme.
- An Integrated IT Platform has been developed to provide an end-to-end solution for the paperless loan processing.
- **SIDBI** has been appointed as the implementation agency for scheme administration.

Reference:

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1753651>

Q.48) Ans: a

Exp:

- **Statement 1 is Incorrect:** The idea was first floated by the **Indian PM** in 2018 during the first assembly of the **International Solar Alliance (ISA)**. He had called for connecting solar energy supply across borders. In May 2021, the United Kingdom and India agreed to combine forces of the Green Grids Initiative and the One Sun One

World One Grid initiative and jointly launch GGI-OSOWOG at the COP26 summit being hosted by the UK at Glasgow in November 2021.

- **Statement 2 is correct:** It aims for a **trans-national electricity grid** supplying solar power across the globe. The ministry of new and renewable energy came out with a Request for Proposal (RFP) to hire consultants for converting this idea into policy. Several policy experts cited it as part of India's answer to China's One Belt One Road infrastructure initiative which entails investment in close to 70 countries.

Extra edge by Onlyias

About International Solar Alliance (ISA):

Officially announced during the UN Climate Change Conference in Paris in 2015, the ISA is a partnership of solar-resource rich countries. Currently, **there are 124(80 countries currently signed and ratified) countries that have agreed to be members for ISA**. Most of these are countries with large participation from Africa, South-east Asia and Europe. **Pakistan and China are not a part of ISA.**

Reference:

https://www.business-standard.com/article/current-affairs/one-sun-one-world-one-grid-all-you-need-to-know-about-solar-strategy-120081500417_1.html

Q.49) Ans: b

Exp:

About SC120-LOX:

The Hindustan Aeronautics Limited (HAL) delivered its heaviest semi-cryogenic propellant tank (SC120- LOX) ever fabricated to the Indian Space Research Organization (ISRO). The semi cryo-liquid oxygen (LOX) tank, the first developmental welded hardware, is a part



of the SC120 stage intended for payload enhancement by **replacing the L110 stage** in the existing Mk-III launch vehicle.

Extra edge by Onlyias

In 2020, HAL had delivered the biggest ever cryogenic Liquid Hydrogen tank (C32-LH2) which is 4m in diameter and 8m in length.

As a strategic reliable partner, HAL has been associating with ISRO for India's prestigious space programmes since the last five decades, and has delivered critical structures, tankages, satellite structures for **PSLV, GSLV-MkII and GSLV-MkIII launch vehicles**.

Various new projects like PS2/GS2 integration, Semi-cryo structure fabrication and manufacturing of cryo and semi-cryo engines are being taken up at HAL, for which installation and commissioning of unique infrastructures are nearing completion.

Reference:

<https://www.newindianexpress.com/nation/2021/oct/08/hal-gives-semi-cryogenic-propellant-tank-to-isro-2369017.html>

Q.50) Ans: a

Exp:

· **Statement 1 is Incorrect:** National Aeronautics and Space Administration (**NASA**) has launched 'Lucy', its first mission to explore the Jupiter Trojan Asteroids. A spacecraft has launched from Cape Canaveral on a mission to uncover "**the fossils**" of the Solar System.

· **Statement 2 is Incorrect:** it is a mission to explore the Jupiter Trojan Asteroids. The Lucy probe will head out to the **orbit of Jupiter** to study **two groups of asteroids** that run in swarms ahead of, and behind, the

gas giant. US space agency (Nasa) scientists say the objects are leftovers from the formation of the planets.

· **Statement 3 is correct:** Lucy has been launched in **October 2021** and, with boosts from Earth's gravity, will complete a **12-year journey** to eight different asteroids, a Main Belt asteroid and seven Trojans, four of which are members of "two-for-the-price-of-one" binary systems.

Extra edge by Onlyias

Inspiration for name Lucy:

· There is a famous human fossil from Africa that was nicknamed **Lucy**, which taught us much about where our species came from. And this new Nasa mission takes direct inspiration and the name from that origins story, except the fossils this spacecraft seeks are hundreds of millions of km from Earth, circling the Sun in formation with Jupiter.

About Trojan Asteroids:

· The Trojans orbit the Sun in two loose groups, with one group leading ahead of Jupiter in its path, the other trailing behind.

· Clustered around the two Lagrange points equidistant from the Sun and Jupiter, the Trojans are stabilized by the Sun and its largest planet in a gravitational balancing act. These primitive bodies hold vital clues to deciphering the history of the solar system.

Reference:

https://www.nasa.gov/mission_pages/lucy/overview/index
<https://www.bbc.com/news/science-environment-58927969>

Q.51) Ans: c



Exp:

· **Statement 1 is Incorrect:** A successful launch of the Agni-5, was carried out on October 27, 2021 from APJ Abdul Kalam Island, Odisha. It is capable of striking **targets at ranges up to 5,000 kilometres** with a very high degree of accuracy. This range puts almost the entire China within the missile's target range. Though officially an ICBM needs a missile to have a range of at least 5,500 km, the Agni 5 is India's closest contender for an ICBM, as it can reach countries across other continents, including parts of Africa and Europe.

· **Statement 2 is correct:** It uses a **three-stage solid fuelled engine**.

· **Statement 3 is correct:** It is a Surface-to-Surface Ballistic Missile. The successful test of Agni-5 is in line with India's stated policy to have 'credible minimum deterrence' that underpins the commitment to 'No First Use'.

Extra edge by Only IAS

About ICBM (Intercontinental ballistic missile):

An intercontinental ballistic missile (ICBM) is a missile with a minimum range of 5,500 kilometres (3,400 mi) primarily designed for nuclear weapons delivery (delivering one or more thermonuclear warheads). Similarly, conventional, chemical, and biological weapons can also be delivered with varying effectiveness, but have never been deployed on ICBMs.

Agni-5 range:

Though the government has claimed that it has a maximum range of around 5,000 km, several reports suggest that it can hit targets as distant as 8,000 km. The nuclear capable missile can carry a warhead of around 1,500 kg and has a launch weight of 50,000 kg, making it one of the most potent missiles in the country.

About Agni missile:

India began testing the Agni series of missiles in 1989 with the first test for Agni 1, an Intermediate Range Ballistic Missile, with a range of around 1,000 km. At

that time only the US, the erstwhile Soviet Union, China, France and Israel, had IRBM technology.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1767034>

<https://indianexpress.com/article/explained/agni-ballistic-missile-vs-china-hypersonic-missile-7595125/>

Q.52) Ans: d

Exp:

· **Statement 1 is correct:** Blue hydrogen is produced mainly from natural gas, using a process called steam reforming, which brings together natural gas and heated water in the form of steam. The output is hydrogen, but also carbon dioxide as a by-product. That means carbon capture and storage (CCS) is essential to trap and store this carbon.

· **Statement 2 is correct:** Pink hydrogen is generated through electrolysis powered by nuclear energy. Nuclear-produced hydrogen can also be referred to as purple hydrogen or red hydrogen.

· **Statement 3 is correct:** Yellow hydrogen is a relatively new phrase for hydrogen made through electrolysis using solar power.

Extra edge by Only IAS

· **White hydrogen:** White hydrogen is a naturally-occurring geological hydrogen found in underground deposits and created through fracking. There are no strategies to exploit this hydrogen at present.

· **Grey hydrogen:** Currently, this is the most common form of hydrogen production. Grey hydrogen is created from natural gas, or methane, using steam methane reformation but without capturing the greenhouse gases made in the process.

Reference:

<https://www.nationalgrid.com/stories/energy->



[explained/hydrogen-colour-spectrum](#)

<https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1752915>

Q.53) Ans: b

Exp:

About Digi Saksham

· The Minister for Labour and Employment has launched **DigiSaksham** - a digital skills programme to enhance the employability of youth by imparting digital skills that are required in an increasingly technology driven era.

Aim of DigiSaksham

· This joint initiative with **Microsoft India** is an extension of the Government's ongoing programs to support the youth from rural and semi-urban areas.

· Through DigiSaksham initiative, free of cost training in digital skills including basic skills as well as advanced computing, will be provided to more than 3 lakh youths in the first year.

· The initiative gives priority to the job-seekers of **semi urban areas belonging to disadvantaged communities**, including those **who have lost their jobs due to Covid-19 pandemic**.

Reference: <https://pib.gov.in/PressReleaseSelfframePage.aspx?PRID=1759684>

Q.54) Ans: c

Exp:

- **Statement 1 is correct:** Global wind report 2021 Released by The Global Wind Energy Council (GWEC). As per the report, total global wind power capacity is up to 743GW now. This is helping the globe to avoid over 1.1 billion tonnes of CO₂ annually. It underlines that Governments across the globe must work with a 'climate emergency' approach to eliminate red tape and planning delays. Government must expand grid

infrastructure to scale-up wind power at the required pace.

- **Statement 2 is correct:** Nationally Determined Contributions (NDC) Synthesis Report released by The UN Framework Convention on Climate Change (UNFCCC). In this report, the UNFCCC has updated the NDC till December 31, 2020 for 75 of the 197 Parties to the UNFCCC. These Parties account for 30% of the global greenhouse gas (GHG) emissions.

Q.55) Ans: a

Exp:

About 'Swordfish'

- Swordfish is an Indian long range tracking radar specifically developed to counter ballistic missile threat.
- Swordfish is an acknowledged derivative of the Israeli Green Pine long radar. However, it differs from the Israeli system as it employs Indian Transmit Receive modules, signal processing, computers and power supplies.
- It is also more powerful than the base Green Pine system and was developed to meet India's specific BMD needs. Bharat Electronics Limited, Electronics and Radar Development Establishment is the manufacturer of Swordfish.
- Capabilities:

- Target acquisition and fire control radar for the BMD system.
- Can Guide Exo-atmospheric interceptor missile PAD to hit its target in space at an altitude over 80 km from earth.
- The Swordfish LRTR currently has a range of 600 km (370 mi)-800 km (500 mi) km range and can spot objects as small as a cricket ball (3-inches in diameter), which the DRDO is in the process of upgrading it to 1,500 km as of 2012. These may end up as entirely new



designs as well.

Q.56) Ans: b

Exp:

Nuclear Research Reactor

- Research reactors are nuclear reactors that serve primarily as a neutron source. They are also called non-power reactors, in contrast to power reactors that are used for electricity production, heat generation, or maritime propulsion.
- The neutrons produced by a research reactor are used for neutron scattering, non-destructive testing, analysis and testing of materials, production of radioisotopes, research and public outreach and education. Research reactors that produce radioisotopes for medical or industrial use are sometimes called isotope reactors.
- Reactors that are optimised for beamline experiments nowadays compete with spallation sources.
- Nuclear Research Reactors in India:
 - Apsara: Designed by BARC and built with the assistance of the United Kingdom. It is at Trombay.
 - Cirus: Built with the assistance of Canada and is present at Trombay
 - Zerlina: Indigenously built zero energy experimental reactor in Trombay
 - Kamini: It is the first reactor in the world designed specifically to use U-233 as fuel. Located at IGCAR (Indira Gandhi Centre for Atomic Research) in Kalpakkam.
 - Dhruva: The Dhruva reactor is India's largest research reactor and primary source of weapons-grade plutonium.

Q.57) Ans: d

Exp:

Mechanism of nuclear power generation

- Nuclear power plants heat water to produce steam. The steam is used to spin large turbines that generate electricity. Nuclear power plants use heat produced during nuclear fission to heat water. In nuclear fission, atoms are split apart to form smaller atoms, releasing energy.
- In nuclear power plants, uranium fuel undergoes nuclear fission and generates an enormous amount of heat. The heat makes high-temperature and high-pressure steam that rotates turbine to generate energy
- Moderator reduces the speed of the neutron produced in nuclear fission to facilitate further fission reaction and sustain a chain reaction.
- A control rod controls the power of a nuclear reactor. By inserting control rods, excessive fission is prevented. A reactor containment vessel, made of steel, accommodates a reactor pressure vessel.

Q.58) Ans: a

Exp:

- A hypergolic propellant is either of the two rocket propellants used in a hypergolic rocket engine, which spontaneously ignite when they come into contact. The two propellants are usually termed the "fuel" and the "oxidizer".
- **Statement 1 is correct:** Although hypergolic propellants are difficult to handle, a hypergolic engine is easy to control and very reliable. Hypergolic propellants need no ignition system. Although larger hypergolic engines in some launch vehicles use turbopumps, most hypergolic engines are pressure-fed. A gas, usually helium, is fed to the propellant tanks under pressure through a series of check and safety valves. The propellants in turn flow through control valves into the combustion chamber; there, their instant contact



ignition prevents a mixture of unreacted propellants from accumulating and then igniting in a potentially catastrophic hard start.

• **Statement 2 is incorrect:** Hypergolic propellants ignite spontaneously on contact with each other and require no ignition source. The easy start and restart capability of hypergolic make them ideal for spacecraft manoeuvring systems. They can fire any number of times by simply opening and closing the propellant valves until the propellants are exhausted and are therefore uniquely suited for spacecraft maneuvering.

Q.59) Ans: b

Exp:

• **Statement 1 is incorrect :** Halogens are highly reactive elements and therefore, they do not exist in free state but exist only in combined form. Halogens have the highest electron affinity so they act as a strong oxidizing agent. Their oxidizing power decreases from fluorine to iodine.

• **Statement 2 is correct:** Halogen is derived from Greek words Hals- sea salt , gennao- producer, therefore fluorine, chlorine, bromine are collectively known as halogens as they react with metals to produce salts.

Q.60) Ans: c

Exp:

About Ozone

• Ozone (O₃) is a highly reactive gas composed of three oxygen atoms. It is both a natural and a man-made product that occurs in the Earth's upper atmosphere (the stratosphere) and lower atmosphere (the troposphere). Depending on where it is in the atmosphere, ozone affects life on Earth in either good or bad ways.

Stratospheric ozone is formed naturally through the interaction of solar ultraviolet (UV) radiation with molecular oxygen (O₂). The "ozone layer," approximately 6 through 30 miles above the Earth's surface, reduces the amount of harmful UV radiation

reaching the Earth's surface.

• Where ozone is formed, **peak concentrations usually occur during afternoon hours**, when sunlight is the most intense. However, areas downwind of major sources of VOC(volatile organic compounds) and NO_x(nitrogen oxides) may experience ozone peaks in the afternoon and evening, after wind has carried ozone and its VOC and NO_x precursors many miles from their sources. Thus, high ozone concentrations can occur in remote areas and at various times of day, including during the early evening or night.

• Ozone is used as a germicide and disinfectant, for sterilizing water and for detecting position of double bonds in unsaturated organic compounds. It is also used as a bleaching agent for oils, flour, ivory and delicate fabrics. Its bleaching action is due to its oxidizing action on the organic matter.

Q.61) Ans: c

Exp:

- **Statement 1 is correct :** Metalloids have properties of metals as well as non- metals. Elements belonging to the semimetals are brittle in nature, but their chemical behavior is distinctly non-metallic. They can, and generally do, form alloys with the metals around them, making them stronger and giving them metallic properties. They are also fair conductors of electricity.
- On a standard periodic table, all of these elements can be found in a diagonal region of the p- block, extending from boron at one end to astatine at the other.
- **Statement 2 is correct :** Metalloids and their compounds are used in alloys, biological agents, flame retardants, glasses, optic storage, pyrotechnics, semiconductors and electronics.
- The six commonly recognised metalloids are Boron, Silicon, Germanium, Arsenic, Antimony and Tellurium. Elements less commonly recognised as metalloids include carbon, aluminium, polonium and astatine.



Q.62) Ans: c

Exp:

· **Statement 1 is correct** : Biogas is an energy-rich gas produced by anaerobic decomposition or thermochemical conversion of biomass. Biogas is composed mostly of methane (CH₄), the same compound in natural gas, and carbon dioxide (CO₂). The methane content of raw (untreated) biogas may vary from 40%–60%, with CO₂ making up most of the remainder along with small amounts of water vapor and other gases. Biogas can be burned directly as a fuel or treated to remove the CO₂ and other gases for use just like natural gas. Treated biogas may be called renewable natural gas or biomethane.

· **Statement 2 is correct** : Biogas makes our energy production safer. It has no negative implications on the environment and is one of the cleanest fuels available. Biogas also helps mitigate methane emissions that would have otherwise escaped from landfills or manure lagoons. Using this methane as a fuel dramatically reduces its climate impact by converting it into CO₂, which is up to 34 times less potent as a greenhouse gas. Increased Energy Security—Biogas offsets non-renewable resources, such as coal, oil, and fossil fuel-derived natural gas. Other renewable projects like solar, wind and water depend a lot on the geography of the project location. Biogas production is location independent and the feasibility is dependent only on the feedstock, which is organic waste. Hence biogas projects are feasible irrespective of the terrain and geography.

Q.63) Ans: a

Exp:

Atlantic Meridional Overturning Circulation

· The AMOC is a **large system of ocean currents, like a conveyor belt, driven by differences in temperature and salt content – the water's density**. It is a large system of ocean currents that carry warm water from the tropics northwards into the North Atlantic.

· As warm water flows northwards it cools and some evaporation occurs, which increases the amount of salt. Low temperature and a high salt content make the

water denser, and this dense water sinks deep into the ocean. The cold, dense water slowly spreads southwards, several kilometres below the surface. Eventually, it gets pulled back to the surface and warms in a process called “upwelling” and the circulation is complete.

· If the AMOC collapsed, it would increase cooling of the Northern Hemisphere, sea level rise in the Atlantic, an overall fall in precipitation over Europe and North America and a shift in monsoons in South America and Africa.

Source: <https://www.metoffice.gov.uk/weather/learn-about/weather/oceans/amoc>

Q.64) Ans: b

Exp:

· **Statement 1 is incorrect**: It is a sacred river in southern India that flows through the state of Karnataka to Andhra Pradesh. The ancient name of the river was Pampa. The river is approximately 710 km long, and it drains an area of 72,200 sq km. It is formed by the confluence of two rivers, the Tunga River and the Bhadra River. Both Tunga & Bhadra Rivers originate on the eastern slopes of the Western Ghats.

· **Statement 2 is correct**: The greater part of the Tungabhadra's course lies in the southern part of the Deccan plateau. The river is fed mainly by rain, and it has a monsoonal regimen with summer high water. It's Major tributaries are the Bhadra, the Haridra, the Vedavati, the Tunga, the Varda and the Kumdavathi. It flows in a more or less northwest direction before joining the eastern river Krishna. The Krishna River finally ends into the Bay of Bengal.

Source : <https://hampi.in/tungabhadra-river>

Q.65) Ans: a

Exp:

· **Statement 1 is correct**: NALSA has been constituted under the Legal Services Authorities Act, 1987 to provide free legal services to weaker sections of society. The aim is to ensure that opportunities for securing



justice are not denied to any citizen by reasons of economic or other disabilities. 'Nyaya Deep' is the official newsletter of NALSA. It organises Lok Adalats for amicable settlement of disputes.

- **Statement 2 is incorrect:** The Chief Justice of India shall be the Patron-in-Chief. Second senior-most judge of the Supreme Court of India is the Executive-Chairman. In every State, the State Legal Services Authority has been constituted to give effect to the policies and directions of the NALSA. The State Legal Services Authority is headed by Hon'ble the Chief Justice of the respective High Court who is the Patron-in-Chief of the State Legal Services Authority. In every District, District Legal Services Authority has been constituted to implement Legal Service. The District Legal Services Authority is situated in the District Courts Complex in every District and chaired by the District Judge of the respective district.

Source : <https://nalsa.gov.in/acts-rules/the-legal-services-authorities-act-1987>

Q.66) Ans: b

Exp:

North Eastern Region District SDG Index 2021

- NITI Aayog has released the first edition of the North Eastern Region District SDG Index and Dashboard 2021–22. It is developed by NITI Aayog in collaboration with the Ministry of Development of North Eastern Region (DoNER).
- The index has been developed with the technical inputs from the United Nations Development Programme.
- To measure the performance of the districts of the eight North Eastern States on the Sustainable Development Goals (SDG) and their corresponding targets and ranks the districts based on the same. Eight states include Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura.
- The index is based on NITI Aayog's SDG India Index –the principal and official tool for monitoring progress on the SDGs at the national and State/ Union

Territory levels. The index utilizes 84 indicators that cover 15 Sustainable Development Goals(SDG) across 50 targets.

Key Findings

- East Sikkim district ranks first in the index, followed by districts Gomati and North Tripura in the second position.
- Out of the 103 districts considered for the ranking, 64 districts belonged to the Front Runner category while 39 districts were in the Performer category in the composite score and ranking of districts.
- All districts in Sikkim and Tripura fall in the Front Runner category, and there are no districts in the Aspirant or Achiever categories.

Source: https://www.business-standard.com/article/economy-policy/niti-aayog-releases-ner-district-sdg-index-report-east-sikkim-tops-chart-121082601260_1.html

Q.67) Ans: c

Exp:

- **Statement 1 is correct:** UNCLOS (United Nations Convention on the Law of the Sea adopted and signed in 1982. It became effective in the year 1994. It replaced the four Geneva Conventions of April, 1958, which respectively concerned the territorial sea and the contiguous zone, the continental shelf, the high seas, fishing and conservation of living resources on the high seas. The Convention has become the legal framework for marine and maritime activities. It divides marine areas into five main zones namely- Internal Waters, Territorial Sea, Contiguous Zone, Exclusive Economic Zone (EEZ) and the High Seas.
- **Statement 2 is correct:** UNCLOS is the only international convention which stipulates a framework for state jurisdiction in maritime spaces. It provides a different legal status to different maritime zones.
The Convention has created three new institutions on the international scene:
 - o The International Tribunal for the Law of the Sea.
 - o The International Seabed Authority.



- o The Commission on the Limits of the Continental Shelf.

Source

: https://www.un.org/depts/los/convention_agreement/s/convention_overview_convention.htm

Q.68) Ans: b

Exp:

It is a standard MOOC (Massive Open Online Course) platform with course material such as filmed lectures and e- study material and Developed by National Institute of Social Defence (NISD) under Ministry of Social Justice and Empowerment (MoSJE). The main objective of introducing courses on social defence is to impart training and enhance the knowledge and skills for the capacity building of the participants

It will provide five basic courses

- Drug (Substance) Abuse Prevention
- Geriatric/Elderly Care
- Care and Management of Dementia
- Transgender Issues
- Comprehensive course on Social Defence Issues.

The courses can be taken up by anyone who wishes to enhance his or her knowledge on the topics and there is no fee for joining.

Source: <https://pib.gov.in/PressReleaseframePage.aspx?PRID=1745781>

Q.69) Ans: a

Exp:

Abanindranath Tagore

- Abanindranath Tagore, the nephew of Rabindranath Tagore, was one of the most prominent artists of the Bengal school of art in India.
- He was the first major supporter of swadeshi values in Indian art. He first created the '**Indian Society of Oriental Art**' and later went on to establish Bengal school of art.
- He believed that Indian art and its art forms gave importance to spirituality as opposed to the West which

stressed on materialism, thus rejecting it.

- His idea of modernizing Mughal and Rajput paintings eventually gave rise to modern Indian painting, which took birth at his Bengal school of art.
- Most of his literary works were meant for children. Some of his books like 'BudoAngla', 'KhirerPutul' and 'Rajkahini' are best examples of Bengali children's literature.

Source: <https://www.culturalindia.net/indian-art/painters/abanindranath-tagore.html>

Q.70) Ans: b

Exp:

'Hunar Haat'

- It is an exhibition of handicrafts and traditional products made by artisans from the minority communities. The exhibition is organised by the Ministry of Minority Affairs under USTTAD (Upgrading the Skills & Training in Traditional Arts/Crafts for Development) scheme.
- Theme of the exhibition was Vocal for Local. These Haat aims to provide market exposure and employment opportunities to artisans, craftsmen and traditional culinary experts.
- It envisages boosting the skills of craftsmen, weavers and artisans who are already engaged in the traditional ancestral work. 'Hunar Haat' has proved to be an "Empowerment Exchange" for master artisans and craftsmen.
- It has proved to be immensely beneficial and encouraging for artisans and craftsmen as lakhs of people visit the "Hunar Haat" and purchase indigenous handmade products of artisans on a large scale.

Source: <https://hunarhaat.org/>

Q.71. Ans: d

Exp:

Sree Narayana Guru

- Sree Narayana Guru was a catalyst and leader from



Kerala who reformed the oppressive caste system that prevailed in society at the time whose philosophy always advocated social equality, education for all, and spiritual enlightenment.

- Born in the Ezhava caste, Narayana Guru had experienced discrimination from the upper caste of society. One of his famous sayings in Malayalam was 'One caste, one religion, one god for all. Narayana Guru consecrated the first temple of Lord Shiva in 1888 where an idol was ordained by a non-brahmin in Aruvippuram village of Kerala sparking off the anti-caste revolution against the upper-caste Brahmin communities.
- Later, in 1903, he established the Sree Narayana Dharma Paripalana Yogam (SNDP) organization as the founder and president that continues to mark its strong presence to this day.
- The famous 'Vaikom Satyagraha' protest movement against caste discrimination and untouchability abolished untouchability and inequality.
- The Sivagiri pilgrimage was established in 1924 to promote the virtues of cleanliness, education, devotion, agriculture, handicrafts, and trade. His philosophy and teachings are a way of life for the people of Kerala.

Source: <https://www.keralatourism.org/varkala/sree-narayana-guru.php>

Q.72) Ans: b

Exp:

miSHERLOCK

- miSHERLOCK is a small tabletop device that can detect SARS-CoV-2 from a saliva sample in about an hour. It is as accurate as PCR tests. It is based on SHERLOCK, a CRISPR-based tool. The device can also be used to detect specific viral mutations linked to some of the variants now circulating.

Steps involved are

- A pre-processing step disables enzymes called salivary nucleases, which destroy nucleic acids such as RNA. Once the sample goes into the device, the nucleases are inactivated by heat and two chemical reagents.

- viral RNA is extracted and concentrated bypassing the saliva through a membrane.

Source: <https://news.mit.edu/2020/sherlock-based-one-step-test-provides-rapid-sensitive-covid-19-detection-0505>

Q.73) Ans: c

Exp:

- **Statement 1 is correct:** Eutelsat Quantum, the world's first commercial fully re-programmable satellite lifted off from French Guiana. Unlike conventional models that are designed and "hard-wired" on Earth and cannot be repurposed once in orbit, Eutelsat Quantum allows users to tailor it almost in real-time.
- **Statement 2 is correct:** This means the satellite can be used to provide mobile coverage for moving objects such as aircraft and ocean going vessels, or coverage after a natural disaster or for one-off events. It has been developed under an ESA (European Space Agency) partnership project with Airbus.

Source : <https://www.hindustantimes.com/world-news/worlds-first-commercial-fully-re-programmable-satellite-lifts-off-here-s-all-you-need-to-know-101627694453032.html>

Q.74) Ans: a

Exp:

- **Statement 1 is correct:** The Sovereign Gold Bond Scheme was launched by Govt in November 2015, under the Gold Monetisation Scheme. Under the scheme, the issues are made open for subscription in tranches by RBI in consultation with GOI. To be issued by Reserve Bank India on behalf of the Government of India. The Bonds will be denominated in multiples of gram(s) of gold with a basic unit of 1 gram. The tenor of the Bond will be for a period of 8 years with exit option in 5th, 6th and 7th year, to be exercised on the interest payment dates.



· **Statement 2 is incorrect:** Minimum permissible investment will be 1 gram of gold. Bonds will be tradable on stock exchanges within a fortnight of the issuance on a date as notified by the RBI. The Gold Bonds will be issued as Government of India Stocks under Government Security Act, 2006. The investors will be issued a Holding Certificate for the same. The Bonds are eligible for conversion into Demat form.

Source :

<https://sbi.co.in/web/personal-banking/investments-deposits/govt-schemes/gold-banking/sovereign-gold-bond-scheme-sgb>

Q.75) Ans: a

Exp:

· **Statement 1 is correct:** The year 2021 will mark the 100th year anniversary of the Malabar rebellion, which is also known as the Moplah (Muslim) riots. It had been an uprising of Muslim tenants against British rulers and local Hindu landlords. It has often been perceived as one of the first nationalist uprisings in southern India.

· **Statement 2 is incorrect:** It has even been described as a peasant revolt. In fact, in 1971, the then Kerala government had included the participants of the rebellion in the category of freedom fighters. The incidents of the uprising took place in regions which are currently under the Malappuram district in north Kerala.

Source: <https://indianexpress.com/article/explained/malabar-rebellion-of-1921-explained-7462838/>

Q.76) Ans: a

Exp:

· **Statement 1 is correct:** The Members of Parliament Local Area Development Division is entrusted with the responsibility of implementation of Members of Parliament Local Area Development Scheme (MPLADS). Under the scheme, each MP has the choice to suggest to the District Collector for works to the tune of Rs.5 Crores per annum to be taken up in his/her constituency. The Rajya Sabha Members of Parliament

can recommend works in one or more districts in the State from where he/she has been elected.

· **Statement 2 is incorrect:** The Nominated Members of the Lok Sabha and Rajya Sabha may select any one or more Districts from any one State in the Country for implementation of their choice of work under the scheme. The Ministry has issued the guidelines on MPLADS Scheme including implementation and monitoring of the scheme. The Department has initiated all necessary steps to ensure that the scheme is successfully implemented in the field.

Source: <https://www.thehindu.com/news/national/50-funds-allotted-for-ongoing-mplads-projects-lapse/article35731768.ece>

Q.77) Ans: a

Exp:

· The Committee on Public Accounts is constituted by Parliament each year for examination of accounts showing the appropriation of sums granted by Parliament for expenditure of Government of India, the annual Finance Accounts of Government of India, and such other accounts laid before Parliament as the Committee may deem fit such as accounts of autonomous and semi-autonomous bodies.

· The Committee on Public Accounts is the oldest Parliamentary Committee and was first constituted in 1921. The Committee consists of 22 Members, 15 Members are elected by Lok Sabha and 7 Members of the Rajya Sabha are associated with it. The Speaker is empowered to appoint the Chairman of the Committee from amongst its members.

· The Committee on Public Accounts scrutinizes the Appropriation Accounts of the Government of India and the reports of the Comptroller and Auditor General of India.

Source: <http://164.100.47.194/loksabha/writereaddata/RTI/righttoinformationact/publicaccountscommitte.htm>



Q.78) Ans: a

Exp:

· **Statement 1 is correct:** A whip is an official of a political party who acts as the party's 'enforcer' inside the legislative assembly or house of parliament. In India, every major political party appoints a whip who is responsible for the party's discipline and behaviour on the floor of the House. There are some cases such as Presidential elections where whips cannot direct a Member of Parliament (MP) or Member of Legislative Assembly (MLA) to vote in a particular fashion.

· **Statement 2 is incorrect:** India inherited the concept of the whip from the British parliamentary system. A legislator may face disqualification proceedings if she/he disobeys the whip of the party unless the number of lawmakers defying the whip is 2/3rds of the party's strength in the house. Disqualification is decided by the Speaker of the house. There are some cases such as Presidential elections where whips cannot direct a Member of Parliament (MP) or Member of Legislative Assembly (MLA) to vote in a particular fashion.

Source: https://www.business-standard.com/article/politics/explained-what-is-whip-in-indian-politics-and-what-does-it-do-what-happens-if-it-s-disobeyed-119112600362_1.html

Types of whips

- One-line whip is issued to inform members of a party about a vote. It allows a member to abstain in case they decide not to follow the party line.
- A Two-line whip is issued to direct the members to be present in the House at the time of voting.
- Three-line whip is issued to members directing them to vote as per the party line.

Q.79) Ans: b

Exp:

· **Adjournment sine die:** Adjournment sine die means

terminating a sitting of Parliament for an indefinite period, that is, when the House is adjourned without naming a day for reassembly, it is called adjournment sine die. The power of adjournment sine die lies with the presiding officer of the House.

· **Adjournment:** An adjournment results in the suspension of work in a sitting for a specified time, which may be hours, days or weeks. In this case, the time of reassembly is specified as an adjournment only terminates a sitting and not a session of the House. The power of adjournment lies with the presiding officer of the House.

· **Prorogation:** The term prorogation means the termination of a session of the House by an order made by the President under Article 85(2)(a) of the Constitution. The prorogation terminates both the sitting and session of the House and is usually done within a few days after the House is adjourned sine die by the presiding officer. The President issues a notification for the prorogation of the session. However, the president can also prorogue the House while in session. It must be noted that all pending notices except those for introducing bills lapse.

· **Dissolution:** Whenever a dissolution happens, it ends the very life of the existing House and a new House is constituted after the General Elections. However, only the Lok Sabha is subject to dissolution as the Rajya Sabha, being a permanent House, is not subject to dissolution.

Source: <https://www.thehindu.com/news/national/parliament-proceedings-lok-sabha-adjourns-sine-die-two-days-before-schedule-registers-22-per-cent-productivity/article35850141.ece>

Q.80) Ans: d

Exp:



Vehicle scrappage policy

- The policy aims to scrap old, unfit and polluting vehicles by creating an infrastructure for automated testing of vehicles that have completed the registration period.
- As per law, a registration certificate for a passenger vehicle is valid for 15 years from date of issue. For a commercial vehicle, it is valid for a period of 10 years.
- The scrappage policy will come into effect after the expiry of this 10 or 15 year period. The vehicle will then have to undergo a mandatory fitness test.
- The Centre has announced several incentives that will be offered to people for retiring their old and unfit vehicles.
- Owners of such vehicles will get a scrap value which will be equivalent to 4 per cent to 6 percent of the ex-showroom price of the new vehicle that they would be purchasing and there will be zero registration fees for new vehicles purchased if the owner shows a certificate of deposit.
- State governments have been asked to offer concessions on motor vehicle tax. The concessions include up to 25 per cent for non-transport vehicles and up to 15 per cent for transport vehicles. Vehicle manufacturers have been advised to provide a 5 per cent discount on purchase of new vehicles against certificate of deposit.

https://www.business-standard.com/article/economy-policy/vehicle-scrappage-policy-will-promote-circular-economy-pm-modi-121081300897_1.html

Q.81) Ans: b

Exp:

PM-KUSUM

- PM-KUSUM (Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan) Scheme is aimed at ensuring energy security for farmers in India, along with honouring India's commitment to increase the share of installed capacity of electric power from non-fossil-fuel sources to 40% by 2030 as part of Intended Nationally Determined Contributions (INDCs).

- The scheme aims to add solar and other renewable capacity of 25,750 MW by 2022 with total central financial support of Rs. 34,422 Crore including service charges to the implementing agencies. State Nodal Agencies (SNAs) of MNRE will coordinate with States/UTs, Discoms and farmers for implementation of the scheme.

The PM-KUSUM Scheme was launched in 2019 with 3 components:

- **Component-A:** For Setting up of 10,000 MW of Decentralized Grid Connected Renewable Energy Power Plants on barren land.
- **Component-B:** For Installation of 17.50 Lakh stand-alone solar agriculture pumps.
- **Component-C:** For Solarisation of 10 Lakh Grid Connected Agriculture Pumps.

<https://www.india.gov.in/spotlight/pm-kusum-pradhan-mantri-kisan-urja-suraksha-evam-utthaan-mahabhiyan-scheme>

Q.82) Ans: c

Exp:

- **Statement 1 is correct:** Zika virus disease is a mosquito-borne disease caused by Zika virus. Most infections are either asymptomatic or cause a mild illness with a transient maculopapular rash. The Aedes aegypti mosquito is the main vector but other Aedes species can also transmit the virus. Viral circulation and a few outbreaks were documented in tropical Africa and in some areas in south-east Asia. Zika virus disease is considered an emerging infectious disease.

- **Statement 2 is correct:** Sexual transmission has been reported recently. According to WHO, there is scientific consensus that Zika virus is a cause of microcephaly, congenital nervous system malformations and Guillain-Barré syndrome. There is no prophylactic or curative treatment or vaccine to protect against Zika virus infection. Therefore, preventive personal measures are recommended to avoid mosquito bites during the daytime.



Source: <https://www.hindustantimes.com/india-news/centre-rushes-team-to-maharashtra-after-first-case-of-zika-virus-found-101627900826466.html>

Q.83) Ans: d

Exp:

· **Statement 1 is Incorrect:** Britain and India introduced a plan to improve connections between the world's electricity power grids to accelerate the transition to greener energy. The "**Green Grids Initiative**" at the **COP26 climate talks in Glasgow, Scotland**, was backed by **more than 80 countries** and could set a model for how rich countries help poorer ones to reduce their emissions and meet the goal of capping global warming at 1.5 degrees Celsius (2.7 Fahrenheit) above pre-industrial norms.

· **Statement 2 is Incorrect:** Its aim is to accelerate the construction of the new generation and transmission infrastructure for a world powered by renewable energy. There is more than enough clean energy to power the world economy, if we build the right grids. A tiny fraction of the world's deserts, the equivalent of a **square 400 km X 400 km (250 miles X 250 miles)**, covered with solar power stations, could produce all the electricity the world uses today. Wind power has similar potential. Existing hydro dams, together with batteries, can help to balance fluctuations in wind and sun. Linking the grids would allow parts of the world with excess renewable power to send it to areas with deficits. For instance, countries where the sun has set could draw power from others still able to generate solar electricity.

Reference:

<https://www.climateparl.net/greengridsinitiative>
https://a5a850c1-2915-4b8e-80e0-17ccf27092dc.filesusr.com/ugd/67825b_a48f9eca3ca245898b15343b87a5603f.pdf

Q.84) Ans: b

Exp:

· **Statement 1 is Incorrect:** Methane is the second-most abundant greenhouse gas in the atmosphere, after carbon dioxide, and, therefore, pledges related to cutting down its emissions are significant. According to the latest **Intergovernmental Panel on Climate Change report**, **methane accounts for about half of the 1.0 degrees Celsius net rise** in global average temperature since the pre-industrial era.

· **Statement 2 is correct:** The Global Methane Pledge was launched at the UN COP26 climate conference in Glasgow. So far, over 90 countries have signed this pledge.

· **Statement 3 is Incorrect:** The pledge was first announced in September by the US and EU, and is essentially an agreement to reduce global methane emissions. One of the central aims of this agreement is to cut down methane emissions by up to 30 per cent from 2020 levels by the year 2030.

Extra Edge by Onlyias

About Methane gas effect on environment

· According to the UN, 25 percent of the warming that the world is experiencing today is because of methane, a greenhouse gas, which is also a component of natural gas. Because it is a greenhouse gas, its presence in the atmosphere increases Earth's temperature.

· There are various sources of methane including **human and natural sources**. Human sources of methane include **landfills, oil and natural gas systems, agricultural activities, coal mining, wastewater treatment, and certain industrial processes**.

· Rapidly reducing methane emissions is complementary to action on carbon dioxide and other greenhouse gases, and is regarded as the single most effective strategy to reduce global warming in the near term and keep the goal of limiting warming to **1.5 degrees Celsius** within reach.



Reference:

<https://indianexpress.com/article/explained/global-methane-pledge-explained-7605172/>

Q.85) Ans: b

Exp:

· **Statement 1 is Incorrect:** The Climate Equity Monitor provides an online dashboard for assessing, at the international level, equity in climate action, inequalities in emissions, energy and resource consumption across the world, and ongoing climate policies of several countries. It is aimed at monitoring the performance of **Annex-I Parties** under the UNFCCC (developed countries) based on the foundational principles of the Climate Convention, namely equity and the principle of common but differentiated responsibilities and respective capabilities (CBDR-RC). The performance and policies of the Non Annex-I Parties (developing countries) will be also provided for comparison.

· **Statement 2 is correct:** The website has been conceptualized and developed by independent researchers from India -- the Climate Change Group at the **M.S Swaminathan Research Foundation (MSSRF)**, Chennai, and the **Natural Sciences and Engineering department at the National Institute of Advanced Studies (NIAS)** Bengaluru, with other independent researchers. The MSSRF team was led by Prof. T. Jayaraman, Senior Fellow, Climate Change and the NIAS team by Dr. Tejal Kanitkar, Associate Professor. The developers of this dashboard have taken this initiative to build awareness, especially among the public of the global South, that climate action is a global collective action problem. The website is intended to debunk the narrative provided by many developed countries, and global non-government organizations that focus attention continually on what developing countries must do, constantly demanding greater commitment and action from them.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1768174>

Q.86) Ans: d

Exp:

· **Statement 1 is correct:** The High Ambition Coalition (HAC) for Nature and People is an intergovernmental group of **now more than 70 countries co-chaired by Costa Rica and France and by the United Kingdom as Ocean co-chair**, championing a global deal for nature and people.

· **Statement 2 is correct:** The central goal of protecting at least 30 percent of the world's land and ocean by 2030. **The 30x30 target is a global target** which aims to halt the accelerating loss of species, and protect vital ecosystems that are the source of our economic security.

· **Statement 3 is correct:** HAC members currently include a mix of countries in the global north and south; European, Latin American, Africa and Asia countries are among the members. India is the **first of the BRICS bloc** of major emerging economies (Brazil, Russia, India, China and South Africa) to join the HAC.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1761855>

<https://www.hacfornatureandpeople.org/why-30x30>

Q.87) Ans: b

Exp:

· **Statement 1 is Incorrect:** India and the United States of America (USA) launched the "Climate Action and Finance Mobilization Dialogue (**CAFMD**)". The dialogue will not only strengthen **India-US** bilateral cooperation on climate and environment but will also help to demonstrate how the world can align swift climate action with inclusive and resilient economic development, taking into account national circumstances and sustainable development priorities. The launch was preceded by a bilateral meet where both sides discussed at length a wide range of climate issues relating to **COP26, Climate Ambition, Climate Finance, Global Climate Initiatives** including



International Solar Alliance (ISA), Agriculture Innovation Mission for Climate (AIM4C).

· **Statement 2 is Correct:** The CAFMD is one of the **two tracks of the India-U.S. Climate and Clean Energy Agenda 2030 partnership** launched at the **Leaders' Summit** on Climate in April 2021, by Prime Minister Shri Narendra Modi and US President Mr. Joseph Biden.

Reference:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1754590>

<https://mea.gov.in/bilateral-documents.htm?dtl/33821/IndiaUS+Joint+Statement+on+Launching+the+IndiaUS+Climate+and+Clean+Energy+Agenda+2030+Partnership>

Q.88) Ans: b

Exp:

· **Statement 1 is Incorrect:** Speaking at the UN climate conference in Glasgow, Prime Minister Narendra Modi shortened the timeline on the country's existing climate targets, and announced a few new targets as well. Modi's announcements underline India's commitment to the fight against climate change — and importantly, none of the targets are likely to be too difficult to achieve. **PM had announced Five-point targets- First is to achieve net zero emission by the year 2070.**

· **Statement 2 is Correct:** In the NDC, submitted in 2015, India said it would ensure that at least 40 per cent of its installed capacity of electricity generation in the year 2030 would come through non fossil-fuel based energy sources. Target has been enhanced by PM. The share of renewables in installed electricity capacity has been increased to 50 per cent.

· **Statement 3 is Correct:** The target of achieving installation of 450GW renewable energy by 2030 has been also increased to 500GW. The **emissions intensity reduction** target has been **raised to 45 percent (earlier it was 33 to 35%)**. The **fifth target** announced by the PM was new. The PM said India would ensure that it reduces **1 billion tonnes of emissions** from its projected

emissions between now and 2030.

Reference:

<https://indianexpress.com/article/explained/india-announcement-climate-targets-7608525/>

Q.89) Ans: c

Exp:

· **Statement 1 is Correct:** Sangai, also called the brow-antlered deer or the dancing deer, is found only in the northeastern state of Manipur, India. It lives in the marshy wetland in the **Keibul Lamjao National Park (KLNP)**, the only floating park in the world. The KLNP is an integral part of the Loktak Lake and is characterized by many floating decomposed plant materials locally called phumdis. The **scientific name of Sangai is Rucervus eldii eldii McClelland.**

· **Statement 2 is Correct:** Sangai is listed as **"Endangered"** in the International Union for Conservation of Nature Red List of Threatened Species and **Schedule I** of the Indian Wildlife (Protection) Act, 1972. Sangai is also included in **Appendix 1** of the Convention on International Trade in Endangered Species of Wild Fauna and Flora.



Extra edge by Onlyias

About Sangai deer

· Sangai or the Indian Eld's deer is one of the three subspecies of Eld's deer found in South and South-East Asia. It has adapted itself to a unique habitat of the floating meadows or phumdi at the national park, a marked difference from the other two, according to background information provided by the Wildlife Institute of India. The sangai is identified under the Ministry of Environment, Forest and Climate Change funded Endangered **Species Recovery Programme** that is **executed by the Wildlife Institute of India**.

· The deer was thought to have gone extinct in 1951 until a remnant population was rediscovered in 1953 at the south-eastern fringe of **Manipur's Loktak lake in a survey conducted under the auspices of the IUCN**.

· The Wildlife Institute of India had called for having a second home for the sangai as the present population at Keibul Lamjao National Park is a single, isolated population, and highlighted that the proposed reintroduction to a second site needs to be expedited.

Reference:

<https://scroll.in/article/991380/in-manipur-the-endangered-sangai-deers-habitat-is-facing-threat-from-climate-change>
https://zeenews.india.com/news/eco-news/sangai-10-interesting-things-you-must-know-about-the-dancing-deer_1891350.html

Q.90) Ans: b

Exp:

About CLAP Initiative:

· Andhra Pradesh chief minister YS Jagan Mohan Reddy launched the **Clean Andhra Pradesh (CLAP)** Jagananna Swachha Sankalpam by flagging off **2,600 garbage collection** vehicles in the state's Vijayawada city.

· The CLAP initiative has been introduced by the state government to bring in a robust sanitation system across rural and urban localities of Andhra Pradesh. The objective of the initiative is to achieve total source segregation of garbage, onsite waste treatment, encouraging home composting, mechanised door-to-door collection with community participation and ensuring complete treatment of waste generated.

· A three-bin system will be set up for primary segregation of waste at a domestic level. Each household in the state will get **blue, green, and red dustbins** to segregate their waste. With regard to door-to-door collection, nearly **3,100 diesel auto tippers and 1,800 electric vehicles** will be deployed. Both these vehicles will have **wet, dry and domestic partition dust bins with a hydraulic lift**.

Reference:

<https://www.hindustantimes.com/india-news/cm-jagan-reddy-to-launch-clean-andhra-pradesh-initiative-today-101633141193165.html>

Q.91) Ans: c

Exp:

· **Statement 1 is Correct:** India does not have any legislation to protect geological sites. The Government of India has attempted to address these concerns on some occasions. Perhaps the most important of them was to introduce the **National Commission for the Heritage Sites Bill on February 26, 2009**. The Bill was withdrawn by the Government from the Rajya Sabha on 31.07.2015.

· **Statement 2 is Correct:** The **UNESCO's Executive Board approved the designation of 8 new sites demonstrating the diversity of the planet's geology as new Geoparks**. With this year's additions, the number of sites in the Global UNESCO Geoparks Network is brought to **169 in 44 countries**. But India does not have



even a single UNESCO designated geopark.

<https://en.unesco.org/global-geoparks>

Extra edge by Onlyias

About UNESCO Geopark:

· UNESCO Global Geoparks are single, unified geographical areas where sites and landscapes of international geological significance are managed with a holistic concept of protection, education and sustainable development. Their bottom-up approach of combining conservation with sustainable development while involving local communities is becoming increasingly popular.

At present, there are **169 UNESCO Global Geoparks in 44 countries**.

· UNESCO Global Geoparks empower local communities and give them the opportunity to develop cohesive partnerships with the common goal of promoting the area's significant geological processes, features, periods of time, historical themes linked to geology, or outstanding geological beauty.

About Global Geoparks Network (GGN):

· The Global Geoparks Network (GGN), of which membership is obligatory for UNESCO Global Geoparks, is a legally constituted not-for-profit organisation.

· The **GGN was founded in 2004** and is a dynamic network where members exchange ideas of best practise to raise the quality standards of a UNESCO Global Geopark.

· While the GGN as a whole comes together **every two years**, it functions through the operation of regional networks, such as the European Geoparks Network that meets twice a year to develop and promote joint activities. It is managed under the UNESCO's Ecological and Earth Sciences Division.

Q.92) Ans: d

Exp:

· **Statement 1 is Correct:** The National Green Tribunal has been **established on 18.10.2010 under the National Green Tribunal Act 2010** for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto. It is a specialized body equipped with the necessary expertise to handle environmental disputes involving multi-disciplinary issues. The Tribunal **shall not be bound by the procedure laid down under the Code of Civil Procedure, 1908, but shall be guided by principles of natural justice**.

· **Statement 2 is Correct:** Supreme Court ruled that the "National Green Tribunal (NGT) is vested with suo motu power in discharge of its functions under the NGT Act". SC observed that "with the distinct role envisaged for it, can hardly afford to remain a mute spectator when no one knocks on its door".

· **Statement 3 is Correct:** The Tribunal's dedicated jurisdiction in environmental matters shall provide speedy environmental justice and help reduce the burden of litigation in the higher courts. The Tribunal is mandated to make and endeavour for disposal of applications or appeals finally within 6 months of filing of the same. Initially, the NGT is proposed to be set up at five places of sittings and will follow circuit procedure for making itself more accessible. New Delhi is the Principal Place of Sitting of the Tribunal and Bhopal, Pune, Kolkata and Chennai shall be the other four places of sitting of the Tribunal.

Reference:

<https://www.thehindu.com/opinion/op-ed/protecting-indias-natural-laboratories/article36951846.ece>

Reference:

<https://indianexpress.com/article/india/cant-afford-to-be-a-mute-spectator-supreme-court-upholds-suo-motu-power-for-national-green-tribunal-7559058/>



<https://greentribunal.gov.in/about-us>

Q.93) Ans: c

Exp:

- **Statement 1 is correct** : WHO has periodically issued health-based air quality guidelines (AQG) to assist governments and civil society to reduce human exposure to air pollution. The present guidelines are applicable to both outdoor and indoor environments. However, these guidelines do not cover occupational settings, owing to the specific characteristics of the relevant exposures and risk reduction policies.
- **Statement 2 is not correct** : AQGs are not legally-binding, like other WHO guidelines.
- **Statement 3 is correct** : The move does not have an immediate effect in India as the National Ambient Air Quality Standards do not meet the WHO's existing norms.

The government has a dedicated **National Clean Air Programme** that aims for a **20% to 30% reduction in particulate matter** concentrations by **2024** in 122 cities, keeping 2017 as the base year for the comparison of concentration.

Q.94) Ans: a

Exp:

ABOUT SAATH INITIATIVE:

- This was recently inaugurated by the Lieutenant Governor of J&K. It will stress on mentoring and market linkages of products created by women associated with SHGs. This initiative aims to impart skills and convert such women's businesses into Higher Order Enterprises.

EXTRA EDGE BY ONLY IAS:

STATEMENT B: It is related to the

Shoonya **Campaign**. The campaign aims to accelerate adoption of Electric Vehicles (EVs) in the urban deliveries segment and create consumer awareness about the benefits of zero-pollution delivery.

STATEMENT C: It is related to the eSAGU initiative of IIIT Hyderabad. It aims to provide expert suggestions to the farmers. With the use of the internet and audio-visual communication facility, the advice is provided to the farmers regularly regarding the various techniques to increase farm productivity

STATEMENT D: Mission Pehal, Under it, Army officers conduct face-to-face interactions with the Kashmiri youth to encourage the youth to express their grievances and the reasons they may have for distrusting the Indian state; the Army officers

Reference:

[J-K LG Sinha inaugurates 'Saath' initiative for women associated with SHGs | Business Standard News \(business-standard.com\)](#)

<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1755119>

[eSagu to the rescue of farmers - The Hindu](#)

Q.95) Ans: b

Exp:

Snow Leopards

- Snow Leopards are vulnerable species that are found in **12 countries—including India, China, Bhutan, Nepal, Pakistan, Afghanistan, Mongolia and Russia**.
- It's scientific name is *Panthera unica*.
- The species being the apex predators, are an indicator for the health of high altitude habitats.



- In India, they are found along the states of Jammu & Kashmir, Ladakh, Uttarakhand, Himachal Pradesh, Sikkim, Arunachal Pradesh.
- Hemis National Park, in Ladakh is regarded as the Snow Leopard capital of the world.
- In addition to that, They are also found in good numbers in **Uttarakhand's Nanda Devi Biosphere Reserve, Gangotri National Park, Askot Wildlife Sanctuary** and other places of altitude between 3000-4500 meters.

NORTHERN RIVER TERRAPIN: They are **Brackish** water species and one of the largest turtles to be found in Southeast Asia. Earlier, they were found in the river mouths of Odisha and the Sunderbans. Described as the world's second most endangered turtle, classified as **critically endangered** in the IUCN red list.

MANDA BUFFALO: Manda buffalo has been recently recognised as the 19th **unique breed** of buffaloes found in India by the **National Bureau of Animal Genetic Resources (NBAGR)**. They are found in the Eastern Ghats and plateau of Koraput region of **Odisha**. The Manda are considered resistant to parasitic infections and are less prone to diseases and can thrive on modest resources.

KASHMIR STAG: **Hangul** or Kashmir stag is the state animal of Jammu and Kashmir and the only surviving red deer in India. They are restricted to the **Dachigam National Park** and are placed in schedule 1 of Wildlife protection act, 1972. IUCN status : **Critically Endangered**.

Reference : [Snow Leopard | Species | WWF \(worldwildlife.org\)](#)

[Odisha: Koraput's Manda buffalo gets unique, indigenous tag - The Hindu](#)

[Indian biologist wins global award in turtle conservation - The Hindu](#)

Q.96) Ans: a

Exp:

Renewable Energy Certificates(RECs)

- A REC, also known as a renewable energy loan, is a market-based instrument in which the owner of a REC can legally claim to purchase renewable energy.
- RECs are also known as green tags, tradable renewable certificates (TRCs), renewable electricity certificates, or renewable energy credits.
- 1 rec is assumed to be equal to 1 MW, so statement 2 is false.
- These are traded on the power exchanges within the forbearance price and floor price prescribed by the Central Electricity Regulatory Commission (CERC) from time to time.
- National Load Despatch Center (NLDC) is responsible for issuance of REC, registration of renewable energy generation facilities etc.
- It comes under the Ministry of Power, (not under the Ministry of New and Renewable Energy) and the validity of a REC is now unlimited unless it is sold (recently, the validity of a REC is 3 years).

Q.97) Ans: a

Exp:

About e-RUPI

- e-RUPI is a digital voucher which a beneficiary gets on his phone in the form of an SMS or QR code. It is a pre-paid voucher, which can be redeemed at any centre that accepts it.
- e-RUPI has launched by the National Payments Corporation of India (NPCI), which oversees the digital payments ecosystem in India.
- NPCI has partnered with 11 banks for e-RUPI transactions. They are Axis Bank, Bank of Baroda,



Canara Bank, HDFC Bank, ICICI Bank, Indian Bank, IndusInd Bank, Kotak Mahindra Bank, Punjab National Bank, State Bank of India and Union Bank of India.

Advantages of e-RUPI:

- e-RUPI does not require the beneficiary to have a bank account, a major distinguishing feature as compared to other digital payment forms.
- Thus e-RUPI is a one time contactless, cashless voucher-based mode of payment that helps users redeem the voucher without a card, digital payments app, or internet banking access.
- It ensures an easy, contactless two-step redemption process that does not require sharing of personal details either.
- e-RUPI is operable on basic phones also, and hence it can be used by persons who do not own smart-phones or in places that lack internet connection.
- Being a prepaid voucher, e-RUPI would assure real time payments to the service provider.

Q.98) Ans: d

Exp:

· **Statement 1 is incorrect:** Asia Pacific Economic Cooperation or APEC is the premier organization for the Asia and Pacific to support sustainable growth and prosperity in the Asia Pacific region. It aims to build a dynamic and harmonious Asia-Pacific community by championing free and open trade and investment, promoting and accelerating regional economic integration, encouraging economic and technical cooperation, enhancing human security, and facilitating a favorable and sustainable business environment. It is headquartered in Singapore. It comprises 21 member countries from the Asian and the American continents. There are no members in the APEC from Africa.

· **Statement 2 is incorrect:** Russia is a member of the APEC grouping and there is no such criteria for excluding countries extending over more than one continent.

· **Statement 3 is incorrect:** India is not a member of APEC while Thailand is a member of the APEC along with Indonesia, Malaysia, Singapore, and others. Source: <https://www.apec.org/About-Us/About-APEC/Member-Economies>

Q.99) Ans: c

Exp:

· **Pair 1 is incorrect:** Saltwater crocodile is considered as the Earth's largest living crocodile species. It is found in Odisha's Bhitarkanika National Park, the Sundarbans in West Bengal and the Andamans and Nicobar Islands. It is also found across Southeast Asia and northern Australia. Its protection status is:
 o IUCN Red List: **Least Concern**
 o Wild Life (Protection) Act, 1972: **Schedule I**
 o CITES: **Appendix I** (except the populations of Australia, Indonesia and Papua New Guinea, which are included in Appendix II).

· **Pair 2 is correct:** Olive Ridley turtles are the smallest and most abundant of all sea turtles found in the world. They are best known for their unique mass nesting called Arribada, where thousands of females come together on the same beach to lay eggs. They are found in warm waters of the Pacific, Atlantic and Indian oceans. The Odisha's Gahirmatha Marine Sanctuary is known as the world's largest rookery of sea turtles. Their protection status is:
 o IUCN Red List: **Vulnerable**



- o Wildlife (Protection) Act, 1972: **Schedule I.** § Russia
- o CITES: **Appendix I**

· **Pair 3 is correct:** Pygmy Hog is one of the very few mammals that **build its own home**, or **nest**, complete with a 'roof'. It is also an **indicator species**. It thrives in **wet grassland**. Today it is mainly found in a **small population in Assam**. Its protection status is:

- o IUCN Red List: **Endangered**
- o Wildlife (Protection) Act, 1972: **Schedule I.**
- o CITES: **Appendix I**

Reference -

<https://www.thehindu.com/news/national/operation-olivia-to-the-rescue-of-olive-ridleys/article34799480.ece>

Q.100) Ans: b

Exp:

· **Arctic Circle**, parallel, or line of latitude around the Earth, at approximately 66°30' N. Because of the Earth's inclination of about $23\frac{1}{2}^{\circ}$ to the vertical, it marks the southern limit of the area within which, for one day or more each year, the Sun does not set (about June 21) or rise (about December 21).

· The length of continuous day or night increases northward from one day on the Arctic Circle to six months at the North Pole. The countries part of this circle are:

- § USA
- § Canada
- § Finland
- § Denmark
- § Norway
- § Sweden



OnlyIAS Nothing Else

Contact us : info@onlyias.com

Visit : dpp.onlyias.in

Contact : +91-7007 931 912

OnlyIAS